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Asymmetric Reference Price Effects and Dynamic Pricing Policies

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Abstract

We consider a group of frequently purchased consumer brands which are partial substitutes and examine two situations; the first where the group of brands is managed by a retailer, and second where the brands compete in an oligopoly. We assume that demand is a function of actual prices and reference prices, and develop optimal dynamic pricing policies for each situation. In addition to researchers studying pricing strategy, our results may interest retailers choosing between hi-lo pricing and an everyday low price, and manufacturers assessing whether to follow Procter & Gamble's lead and replace a policy of funding consumer price reductions through trade deals with a constant wholesale price.

A reference price is an anchoring level formed by customers based on the pricing environment. The literature suggests that demand for a brand depends not only on the brand price, but also whether the brand price is greater than the reference price (a perceived loss) or is less than it (a perceived gain). The responses to gains and losses are asymmetric. Broadly speaking, we find that when enough consumers weigh gains more than losses, the optimal pricing policy is cyclical. Likewise, when they weigh losses more than gains, a constant price is optimal. Thus, we provide a rationale for dynamic pricing which is quite distinct from the three explanations previously offered: (1) decreasing unit variable costs due to learning effects, (2) the transfer of inventory to consumers who face lower inventory holding costs than do retailers, and (3) competitive effects. Our explanations apply even when the other explanations do not, i.e., in mature product categories where learning effects are minimal, when retailer inventories are minimized through the use of just-in-time policies and when competitive effects do not exist, as in a monopoly.

Greenleaf (1995) has shown numerically that in the presence of reference price effects, the optimal pricing policy for a monopolist can be cyclical. We first analytically extend Greenleaf's result to a monopolist with a constant cost of goods, facing a homogeneous market where all customers either weigh gains more than losses or vice versa. Using this building block we examine a monopolist retailer managing multiple brands. We assume that demand is a linear function of prices of multiple brands, and together with an expression which reflects the reference price effect. Further, we assume that the retailer maximizes average profit per period. Next, we analyze a duopoly and extend the results to an oligopoly. We assume that the manufacturers are able to set the retail prices, as in an integrated channel. Here, we retain the same demand function as for the retailer and derive Markov Perfect Nash equilibria. We use two alternative processes of reference price formation: the exponential smoothed (ES) past price process which is frequently used in the literature, and for the multi-brand situations, the recently proposed reference brand (RB) process (Hardie, Johnson, and Fader 1993). In the latter, the reference price is the current price of the last brand bought—the reference brand. We adapt the individual level RB formulation in Hardie et al., to an aggregate demand specification. For the ES process, we obtain most results analytically; for the RB process we use simulation. Finally, we extend our results to a population with two customer segments: Segment 1 which weighs gains more than losses, and Segment 2 which does the opposite, i.e., is loss averse.

When the market consists exclusively of Segment 1 customers and ES is the reference price process, we find that prices are cyclical in all cases analyzed, i.e., for a monopoly, a monopolist retailer managing multiple brands, a duopoly, and an oligopoly. If the RB formulation is the underlying process, a monopolist retailer managing two brands uses cyclical prices, but in a duopoly, the equilibrium solution is for the brands to maintain constant prices. When all customers belong to Segment 2 (i.e., they are loss averse) constant prices are optimal in all cases for both reference price formulations. When the population consists of both Segment 1 and Segment 2 and the ES process applies, we develop a sufficient condition for cyclical pricing policies to be optimal. The condition is expressed in terms of the proportion of the two segment sizes, the absolute difference between the gain and loss parameters of each segment, and their respective exponential smoothing constants. Interestingly, for reasonable values of the latter two factors, cyclical policies are optimal even when the proportion of Segment 1 is quite small. Similar magnitudes are obtained numerically for the RB case.

(*Reference Prices; Pricing Research; Promotion*)

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1. Introduction

Reference prices are certain anchoring levels for prices that customers form as a function of the pricing environment, for example, past prices. Recently, Greenleaf (1995) has shown that in the presence of reference price effects, the optimal pricing policy for a monopolist is cyclical. Reference price, r, for a brand is assumed to be the exponentially smoothed value of past brand prices. The impact of reference price on demand is based on the difference $(r - p)$, where p is the observed price. When $r > p$, the impact of reference price on demand is $\delta(r - p)$, and its impact on demand when $r < p$ is $\gamma(r - p)$. The parameters δ and γ are interpreted as the gain and loss effects, respectively, since they are associated with perceived gains and losses.

The purpose of this paper is to generalize Greenleaf's results. We begin by considering the pricing decisions of a profit maximizing monopolist with a constant cost of goods. We then extend our analysis to a monopolist retailer managing multiple brands. Next, we analyze a duopoly and extend the results to an oligopoly. In all these scenarios, we use an exponential smoothing process for reference price formation; this is by far the most common mechanism used in the literature (see Greenleaf 1995, Kalwani et al. 1990, Krishnamurthi et al. 1992, Putler 1992, Winer 1985, 1986). We also consider a second reference price formation process, in which the reference price is the current price of the last brand bought—the reference brand. In a recent paper Hardie et al. (1993) have shown that this formulation has empirical validity in a multibrand context. We adapt the individual level reference brand formulation in Hardie et al. (1993) to an aggregate demand specification and determine the optimal pricing policy for a monopolist managing two brands as well as for a duopolist.

We first examine markets in which customers are homogeneous in the way they value gains and losses for each of the brands. In this environment, we study the following three cases for $i = 1, \ldots, N$ brands: (a) $\delta_i > \gamma_i$, (b) $\delta_i \leq \gamma_i$, and (c) $\delta_i > \gamma_i$, $i = 1, \ldots, n_1$ brands and $\delta_i \leq \gamma_i$, $i = n_1 + 1, \ldots, N$ brands. We then extend our analysis to a heterogeneous market where there are two customer segments: Segment 1, which values gains more than losses ($\delta_1 > \gamma_1$); and Segment 2, which values losses more than gains ($\delta_2 \leq \gamma_2$), i.e., Segment 2 is loss averse.

Considering homogeneous customers with $\delta > \gamma$, our analytical findings and simulations support Greenleaf's results. Using the exponential smoothing process of reference price formation, we find that prices are cyclical in all cases, i.e., a monopoly, a monopolist retailer managing multiple brands, a duopoly, and an oligopoly. However, when $\delta \leq \gamma$, our results show that constant prices are optimal. Using the reference brand formulation, we find that when $\delta > \gamma$, the monopolist retailer managing two brands uses cyclical prices, but in a duopoly, the equilibrium solution is to maintain constant prices. When $\delta \leq \gamma$, constant prices again are optimal. We next consider the heterogeneous market and develop a sufficient condition for cyclical pricing policies to be optimal. When reference price is modeled as the exponentially smoothed past price, the condition is expressed in terms of the proportion of the two segment sizes, the absolute difference between the gain and loss parameters of each segment, and their respective exponential smoothing constants. Interestingly, for reasonable values of the latter two factors, cyclical policies are optimal even when the proportion of Segment 1 is quite small. Similar magnitudes are obtained numerically for the reference brand case.

Our results provide a rationale for dynamic pricing which is quite distinct from previously offered explanations. One such explanation involves decreasing unit variable costs. The phenomena that lead to such decreases are the experience curve effect (Robinson and Lakhani 1975, Kalish 1983), and trade promotions offered by the manufacturer (Blattberg and Neslin 1990, Chapter 11). A second explanation concerns the transfer of inventory to consumers who face lower inventory holding costs than do retailers (Blattberg et al. 1981). Finally, competitive effects also explain why prices are dynamic (Lal 1990, Rao 1991).

In a recent strategic move, P&G replaced the long established policy of periodic trade dealing with a policy of constant prices to retailers for some of its brands (*Advertising Age*, March 9, 1992). P&G's actions were widely followed (*Brandweek*, October 26, 1992). In this new environment, our results will assist retailers to answer the following questions: should they abandon the traditional hi-lo pricing (often funded by trade deals) and also adopt a constant pricing policy for consumers? Or, are there conditions under which a hi-lo policy is more profitable?

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The balance of the paper is organized as follows. Section 2 provides a brief review of the relevant literature. Section 3 contains the major results of the paper. In §3.1 we consider a general demand function and develop analytical results for the monopoly case. Simulation is used to determine the effect of the gain and loss parameters on the optimal pricing policy. In §3.2 we extend these results to a monopolist retailer managing multiple brands with interdependent demand. In §3.3 we introduce competition and develop results for a duopoly. These results extend easily to an oligopoly. In §3.4, we consider the reference brand specification for reference price formation (Hardie et al. 1993) in a two brand context. We develop policies for a monopolist retailer and for a duopoly. In §3.5 we consider heterogeneity and develop a sufficient condition for the optimality of cyclical pricing policies. Finally, we provide a discussion in §4.

2. Background

2.1. The Formation of Reference Prices

During the past decade, there have been many theoretical and empirical studies that focus on the formation of reference prices. For a recent review see Kalyanaram and Winer (1995). Among the theoretical works, Kalyanaram and Little (1989) and Winer (1988) provide alternative formulations for the reference price formation process. These formulations stem from theories such as the Weber-Fechner Law in psychophysics, adaptation-level theory, assimilation-contrast theory, prospect theory, and the adaptive expectations framework. Empirical testing of these formulations has been limited. Most of the empirical studies have used the adaptive expectations framework (Nerlove 1958) in modeling consumers' reference price (Greenleaf 1995, Kalwani et al. 1990, Krishnamurthi et al. 1992, Putler 1992, Winer 1985, 1986). According to this framework, reference price is an exponentially weighted average of past observed prices, where more recent prices are given greater weight. Let r_t and p_t be the reference price and the observed price respectively for a brand in period t. Exponentially weighting past prices yields:

$$r_t = \alpha r_{t-1} + (1 - \alpha)p_{t-1}, \quad 0 \leq \alpha < 1. \qquad (1)$$

If $\alpha = 0$, reference price in period t equals observed price in period $t - 1$. As α increases, r_t becomes increasingly dependent on past prices. Thus, α can be regarded as a memory parameter with $\alpha = 0$ corresponding to a one-period memory. Greenleaf (1995) estimates Equation (1) using weekly peanut butter data and finds the best fit when $\alpha = 0.925$. Using scanner panel data on refrigerated orange juice, Hardie et al. (1993) obtain a best fit when $\alpha = 0.83$. Other empirical studies assume a one-period memory, i.e., $\alpha = 0$ (Krishnamurthi et al. 1992, Mayhew and Winer 1992, Winer 1989).

A different approach is proposed and tested by Hardie et al. (1993). They model reference dependence in a multinomial logit framework by considering the last brand purchased as the reference brand. Price gain (loss) occurs if the current price of the target brand is less (greater) than the current price of the reference brand. They find that a model based on the reference brand specification exhibits a better fit than one based on the adaptive expectations framework for the formation of reference prices.

2.2. The Effect of Reference Price on Demand

In the reference price literature, the impact of reference price on demand is modeled as a linear function of the difference between reference price and observed price, a specification which has received strong empirical support. Winer (1985) proposed that according to Prospect Theory (Kahneman and Tversky 1979), the effect of $(r_t - p_t)$ on demand is asymmetric depending on whether it is positive or negative. Prospect Theory predicts that when $(r_t - p_t)$ is positive, consumers perceive a gain; when it is negative, they perceive a loss. Further, the absolute level of the change in demand due to a loss is greater than the corresponding impact of an equal gain. Such behavior results in kinked demand curves; the kinks appear at points where the observed price is equal to the reference price (Putler 1992). But the empirical evidence in this regard has been mixed. In his analysis of market level sales data on eggs, Putler (1992) found that, consistent with Prospect Theory, the effect of a loss on demand is greater than that of an equal gain (2.4:1). Hardie et al. (1993) find a similar result. Greenleaf's (1995) results show that the effect of a gain is greater than that of a loss (5.4:1). On the other hand, using scanner panel data for six brands in two frequently purchased product classes Krishnamurthi et al.

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(1992) find that for brand switchers,[1] the impact on demand of a loss is less than that of an equal gain. For brand loyals, the loss and gain effects are equal. More recent work continues to explore the idea that there may be heterogeneity among consumers both in the reference price formation process and in the parameters of the process itself. Thus, customers may use a variety of heuristics to judge whether a brand is a bargain. Even if all customers use a single process, their parameter values may differ. For example, customers may vary in their memory parameter α, and in their gain and loss parameters δ and γ. Hence, customers can have heterogeneous reference prices (due to different memory parameters) and may exhibit different responses to price changes (due to different gain and loss parameters). Examples of such work are Bell and Lattin (1993), Briesch et al. (1994), Ghose and Lowengart (1994), and Helsen and Hardie (1994).

2.3. Models of Pricing Behavior

Dynamic pricing, particularly in the context of price promotions, has been the focus of study for more than two decades. Varian (1980) shows that heterogeneity in information among consumers is a rationale for dynamic prices. Blattberg et al.'s (1981) results suggest that retail prices are dynamic due to the transfer of inventory holding costs to consumers. Raju et al. (1990) show that a brand with lower brand loyalty has more to gain from price promotions. Kinberg et al. (1974) indicate, although not in an equilibrium framework, that premium brands are better off maintaining cyclical prices when faced with a private label brand. Lal (1990) argues that price promotions can be interpreted as a long-run strategic implicit collusion between national firms to defend their market shares from possible encroachment by other firms. In an asymmetric duopoly, Rao (1991) suggests that in equilibrium the national brand promotes to ensure that the private label brand does not encroach on its customers. The implication of this work is that in symmetric duopolies, the equilibrium strategy is not to offer price promotions.[2] It is the existence of an asymmetric brand (i.e., the private label brand) that provides an incentive for the national brand to offer price deals. In this paper, we show that the reference price phenomenon plays an important role in determining whether it is optimal for firms in a symmetric duopoly to offer cyclical prices.

[1] Consumers who did not purchase any one of the brands at least 50% of the time were classified as switchers

[2] This result can be fairly easily established in the case of aggregate linear interdependent demand functions and considering the option of charging either a constant price or a hi-lo price.

3. Model Development

3.1. Monopoly

We consider a monopolist who maximizes the sum of discounted profits over an infinite time horizon for a single brand. The decision variable in each period t is price, p_t, and we consider the set of prices where demand is positive. Therefore, we assume that prices are bounded, i.e., $c \leq p_t \leq b$, where c is the constant unit cost, b is the upper bound for prices such that it is the highest price where demand is positive, $t = 0, 1, 2, \ldots$. Since our emphasis is on the effect of price, we assume that all other factors Z effecting profit are given and held constant. Let $f(p_t; Z)$ be the profit in period t; f is continuous and differentiable in $[c, b]$. For notational simplicity, we omit Z in the rest of this paper. By definition $f(c) = 0$. In the absence of reference price effects the firm's problem is to select a sequence of prices $\{p_t\}$ to maximize the sum of discounted profits; that is, to maximize $\sum_{t=0}^{\infty} \beta^t f(p_t)$, where $0 < \beta < 1$ is the discount factor. Letting $q(p_t)$ represent demand, the monopolist's problem can be written as:

$$\max_{\{p_t\}} \sum_{t=0}^{\infty} \beta^t (p_t - c) q(p_t),$$

$$\text{subject to } c \leq p_t \leq b. \tag{2}$$

The maximization problem in (2) is equivalent to a single-period optimization problem for which the solution is $p_t = p^*$: $p^* = \arg\max f(p)$: $p \in [c, b]$.

We now consider the effect of reference price, r_t. We assume that the reference price formation process is given by Equation (1), and that the reference price effect on demand is given by:

$$h(r_t, p_t) = \begin{cases} \delta(r_t - p_t) & \text{if } r_t > p_t, \\ \gamma(r_t - p_t) & \text{otherwise,} \end{cases} \tag{3}$$

where $\delta, \gamma > 0$.

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In Equation (3), δ is the gain per unit of price difference when the observed price is less than the reference price. Similarly, γ is the loss per unit of price difference when the observed price is greater than the reference price. Following the reference price literature, we assume that demand, d_t, is a function of two additively separable factors, $q(p_t)$ and $h(r_t, p_t)$:

$$d_t = q(p_t) + h(r_t, p_t). \tag{4}$$

The monopolist's decision problem can now be reformulated as follows: given an initial reference price, r_0, select $\{p_t\}$ to maximize $\Sigma_{t=0}^{\infty} \beta^t(p_t - c)d_t$, i.e.,

$$\max_{p_d} \sum_{t=0}^{\infty} \beta^t(f(p_t) + (p_t - c)h(r_t, p_t)),$$

$$\text{subject to } r_t = \alpha r_{t-1} + (1 - \alpha)p_{t-1}, \quad r_0 \text{ given.} \tag{5}$$

We use dynamic programming to solve (5), an optimal control problem in discrete time with reference price as the state variable and price as the control variable. Following Stokey and Lucas (1989) we have the following recursive equation:

$$H(V)(r_t, p_t) = (p_t - c)d_t + \beta V(p_{t-1}, p_{t-2}, \ldots), \quad \text{where} \tag{6}$$

$$T(V)(r_t) = \sup_{\{p_t\}} H(V)(r_t, p_t). \tag{7}$$

The existence and uniqueness of the solution to $T(V) = V$ with and without discounting is shown in Bertsekas (1987, Chapters 5 and 7). Accordingly we have:

$$H(V)(r_t, p_t) = \begin{cases} f(p_t) = \beta V(r_{t+1}) + (p_t - c)\delta(r_t - p_t) & \text{if } r_t > p_t, \\ f(p_t) + \beta V(r_{t+1}) + (p_t - c)\gamma(r_t - p_t) & \text{otherwise,} \end{cases} \tag{8}$$

where $r_{t+1} = \alpha r_t + (1 - \alpha)p_t$, r_0 given.

We first study the case where all customers belong to Segment 1, i.e., value gains more than losses ($\delta > \gamma$). We characterize the pricing policy in Proposition 1 below as a cyclical policy and explore the policy numerically. We find that optimal prices follow a hi-lo pattern, and the hi-lo range increases as δ increases, given γ. We next study the case where all customers belong to Segment 2, i.e., they are loss averse ($\delta \leq \gamma$). We characterize the pricing policy in Proposition 2 for a number of special cases; we find that the optimal policy is a constant price. We further explore this case numerically and find that for all cases examined, a constant price is optimal.

PROPOSITION 1. *For $0 \leq \alpha < 1$, $0 < \beta < 1$, $\delta > \gamma$, if f is continuous and differentiable, or if f is convex, a constant pricing policy is not optimal. This implies that the optimal price cycles.*

The proof of this and all other propositions is in the Appendix.

According to Proposition 1, given a constant unit cost c, when the effect of gains is higher than that of corresponding losses, the monopolist should vary prices over time regardless of the memory parameter α and the discount rate β. When $\delta > \gamma$, the demand with a pricing policy of $(p^{**} + \xi)$ and $(p^{**} - \xi)$ in two consecutive periods is higher than that which would be obtained if price is constant at p^{**} in both periods. Since the corresponding total profits are also higher, a constant price is not optimal.

This result in Proposition 1 is in contrast to earlier studies which suggest that a primary motivation for offering cyclical prices is to prevent encroachment by a lower priced brand (Lal 1990), a situation which is not encountered by a monopolist. Reference price effects induce a time dependence in demand, a phenomenon which has not been considered in the earlier studies. We have shown that by considering these effects, we arrive at policies which can make cyclical prices optimal even in the absence of cost variations, inventory transfer and competition.

We next examine the effect of the model parameters on the optimal prices and the length of the pricing cycle using simulation. We assume the following linear demand function and Equation (1) for the formation of reference price:

$$d_t = a_1 - b'p_t + g(r_t - p_t), \tag{9}$$

where $g = \delta$ if $r_t > p_t$, else $g = \gamma$.

The memory parameter α is varied at five levels, 0, 0.2, 0.4, 0.6, and 0.8. Two levels of the discount rate (β), 1.0 and 0.99, are used to examine the impact of discounting ($\beta = 1$ implies no discounting); γ is varied at two levels, 0.1 and 0.5. The ratios of δ to γ considered

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are 1.5, 2.0, and 2.5.[3] Since our focus is on evaluating the asymmetric effect of reference price, the remaining parameters, a_1, b', and c are held constant at 1.0, 0.2, and 0.5, respectively. Thus, the simulation uses a $5(\alpha)X2(\beta)X2(\gamma)X3(\delta)$ full-factorial design; the parameter levels are selected such that the demand does not exceed the market potential, a_1, obtained when price is equal to zero.

The dynamic programming algorithm is solved using backward induction and interpolation. Future reference prices are truncated at the second decimal place and future optimal profits are then retrieved using interpolation. In any such optimization, transient effects are encountered at the beginning and at the end of the time horizon. We consider a time horizon of 50 periods and analyze the pricing policy in periods 10 through 40 to remove these effects. We follow this approach in all those simulations where we seek to characterize optimal steady state policies. Thus, in each period, given r_t, we search for prices that maximize the following:

$$V_t(r_t) = (p_t - c)(a_1 - b'p_t + g(r_t - p_t)) + \beta V_{t+1}(r_{t+1}),$$

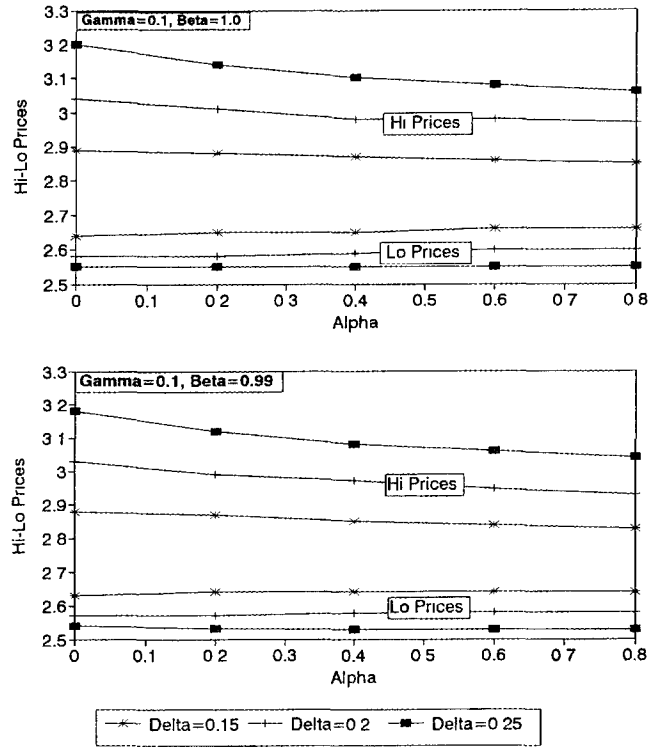
where $g = \delta$ if $r_t > p_t$, else $g = \gamma$, $r_{t+1} = \alpha r_t + (1 - \alpha)p_t$, $V_T(r_{T+1}) = 0$, and $T = 50$.

The one-period optimizing price for the demand function in Equation (9) when $r_t = p_t$ is $p^{**} = 2.75$. We search for prices in the range $[c, b]$ that is $[0.5, 5]$ in steps of 0.01. The optimal prices alternate between a high and a low value. The following result summarizes our finding:

RESULT 1. For $0 \leq \alpha < 1$, $\delta > \gamma$, and demand specified by Equation (9), the optimal pricing policy that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is hi-lo. Further, given γ, the hi-lo range increases with δ.

Figure 1 illustrates Result 1 for the two discount rates used and $\gamma = 0.1$. We obtain similar results for $\gamma = 0.5$. As δ increases, the hi-lo range increases. For a given level of γ, as δ increases, the demand during a hi-lo pricing cycle increases. The monopolist can take advantage of this effect by complementing it by increasing the hi-lo range since the effect of the gain parameter is a function of the price differential. In addition, we find that the hi price is higher for lower levels of the memory parameter α. As α decreases, reference prices become more sensitive to the last period's price; for example, when $\alpha = 0$, current reference price is equal to the last period's price. As the reference price becomes more sensitive to the past period's price, demand during a hi-lo pricing cycle can be increased by increasing the gap between the reference price and actual price. This is achieved by increasing the hi price.

Figure 1 Monopoly: Effect of Alpha, Delta on Hi-Lo Prices



PROPOSITION 2. i. *For $0 \leq \alpha < 1$, $0 < \beta < 1$, $\delta = \gamma$, and for any arbitrary function f, the optimal pricing policy monotonically converges to a steady state price, p^{**}, which is a function of initial reference price r_0. When $r_0 < (>)\ p^{**}$, p_t monotonically increases (decreases) to p^{**}.*

ii. *For $\alpha = 0$, $0 < \beta < 1$, $\delta \leq \gamma$, and for any arbitrary function f, the optimal pricing policy monotonically converges to a steady state price, p^{**}, which is a function of initial reference price r_0. When $r_0 < (>)\ p^{**}$, p_t monotonically increases (decreases) to p^{**}.*

[3] We also consider the ratio 5.4:1 reported in Greenleaf (1995) and found similar results.

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iii. *For $0 < \alpha < 1$, $\delta \leq \gamma$, for any arbitrary function f, and an average profit per period criterion, steady state pricing cycles with periodicity 2 or 3 are not optimal.*

Proposition 2 highlights the importance of the gain/loss parameters in determining optimal pricing strategies. Broadly speaking, when the gain parameter is less than or equal to the loss parameter, cyclical pricing policies are not optimal; constant prices are. When $\delta \leq \gamma$, the impact of a price decrease on demand is less than or equal to that of a corresponding price increase. For example, if a $(p^{**} + \epsilon)$, $(p^{**} - \epsilon)$ pricing policy is followed, the total demand during the two periods will be less than the demand obtained if a constant price of p^{**} is maintained. Note that these results do not require the function f to satisfy any conditions. Proposition 2 leads us to conjecture that:

For $0 \leq \alpha < 1$, $0 < \beta < 1$, $\delta \leq \gamma$, and for any continuous function f, the optimal pricing policy monotonically converges to a constant price, p^{**}, which is a function of initial reference price r_0.

This conjecture extends Proposition 2 to include all values of α, β, and all cases where $\delta \leq \gamma$. We explore it numerically and consider two demand specifications in addition to the linear demand function in Equation (9). These specifications have two components: one for the normal price impact, and one for the impact of reference price. The impact of the price component is captured by negative exponential and log-linear functions, which are widely used in marketing (Lilien et al. 1992):

Demand Specification 1:

$$d_t = a_1 \exp^{-b'_1 p_t} + g(r_t - p_t). \tag{10}$$

Demand Specification 2:

$$d_t = a_1 + a_2 p_t^{-b'_2} + g(r_t - p_t), \tag{11}$$

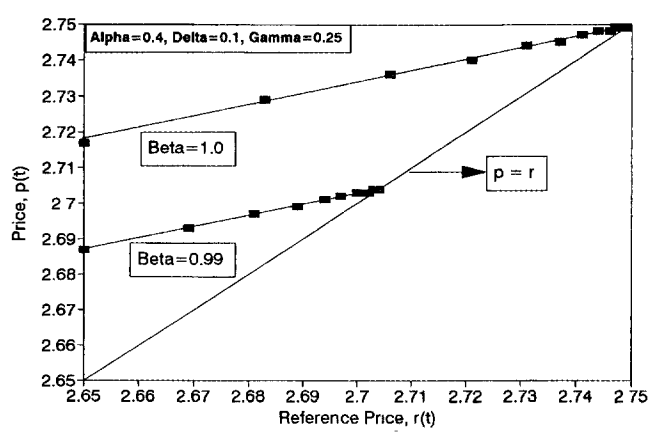
where $g = \delta$ if $r_t > p_t$, else $g = \gamma$.

The memory parameter α is varied at five levels, 0, 0.2, 0.4, 0.6, and 0.8. Two levels of discount rate (β), 1.0 and 0.99, are used; δ is varied at two levels, 0.1 and 0.5. The ratios of γ to δ considered are 1.0, 1.5, 2.0, and 2.5, which include the findings of Putler (1992). Since our focus is on evaluating the asymmetric effect of reference price, the remaining parameters, a_1, b'_1, b'_2, and c are held constant at 1.0, 0.2, 0.2, and 0.5, respectively. Thus, the simulation uses a $5(\alpha)X2(\beta)X2(\gamma)X4(\delta)$ full-factorial design. Note that the parameter levels are selected such that the demand does not exceed the market potential, a_1, obtained when price is equal to zero. The following result summarizes our findings:

RESULT 2. For $0 \leq \alpha < 1$, $\delta \leq \gamma$, and demand specified by Equations (9)–(11), the optimal pricing policy that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, monotonically converges to a constant price. Further, when there is no discounting, the constant pricing policy is equal to the single period profit maximizing price.

Figure 2 presents the result for the linear demand function. We study the transient phase at the beginning of the time horizon since the steady state is a constant price. We report the results for $\alpha = 0.4$, $\delta = 0.1$, $\gamma = 0.25$, $\beta = 0.99$, 1.0. The nature of the results is similar for the other levels of the parameters used in the simulation. When there is no discounting ($\beta = 1.0$) and initial reference price $r_0 = 2.65$, we find that $p_0 = 2.72$ which results in $r_1 = 2.69$, the optimal reaction to which is $p_1 = 2.73$. The resulting reference price is 2.71 leading to $p_2 = 2.74$, and so on. Thus, when r_0 is less than the steady state value 2.75, the one-period optimizing price, the optimal price path monotonically increases to 2.75. When there is discounting ($\beta = 0.99$), the steady state value is 2.71, which is a function of r_0 and the discount rate, and the convergence to the steady state value is more rapid than when there is no discounting.

Figure 2 **Monopoly: Price Path when Delta < Gamma**



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The analytical and numerical results presented so far show that given a constant cost, the monopolist will use a cyclical pricing policy when the gain parameter δ is greater than the loss parameter γ, and maintain a constant price if $\delta \leq \gamma$. The latter result is in contrast to Greenleaf's (1995) result, which suggests that the policy may be cyclical if the hi price is high enough such that the demand is zero, while at the same time assuming that these hi prices are incorporated into the reference price formation process. By restricting our analysis to prices for which demands are positive, we exclude such prices in our analysis.

If a cyclical pricing policy is employed, customers may buy only on deal; thus, demand may go down to zero at the high price. However, customers incur a holding cost (including the cost of spoilage) when they forward buy; this deters them from stocking up. Consequently, there are always new customers entering the market because of stockouts (Blattberg et al. 1981). Therefore, demand will not be zero during off-deal periods. In a comprehensive study of hi-lo versus constant prices in retail supermarkets, Hoch et al. (1994) report that while demand is low during hi periods, it is not zero. As an illustration, consider the pricing policy of Coke, which is widely known to be on deal with an approximately constant periodicity. Although the sales of Coke during off-deal periods are low, they are not zero.

3.2. Monopolist Retailer Managing Multiple Brands

We now consider the case of a monopolist retailer managing multiple brands. The demand for brand i, $i = 1, \ldots, N$, is a function not only of its own price and reference price, but of the prices of all the remaining $(N-1)$ brands. We assume that the reference price of brand i is a function of its own past prices, i.e., customers form expectations for the price of each brand independent of the prices of competing brands. This formulation has been used in empirical work involving multiple brands (for example, see Krishnamurthi et al. 1992). We investigate pricing policies under two scenarios—one where the sum of discounted profits for all the brands over a time horizon is maximized, and one where the average profit per period is maximized. The second approach is more appropriate when evaluating stationary cyclical policies. Extending the function f discussed in §3.1 to multiple brands, the objective function when the retailer maximizes the sum of discounted profits is given by:

$$\max_{\{p_{it}, i=1,N\}} \sum_{i=1}^{N} \sum_{t=0}^{\infty} \beta^t (f_i(p_{1t}, p_{2t}, \ldots, p_{Nt}) + (p_{it} - c) g_i(r_{it} - p_{it})), \quad (12)$$

where $g_i = \delta_i$ if $r_{it} > p_{it}$, else $g_i = \gamma_i$; $\delta_i, \gamma_i > 0$; $r_{it} = \alpha r_{it-1} + (1-\alpha) p_{it-1}$, r_{i0} given.

Thus, we have the following recursive relation:

$$H(V)(r_{1t}, r_{2t}, \ldots, r_{Nt}, p_{1t}, p_{2t}, \ldots, p_{Nt}) = f(p_{1t}, p_{2t}, \ldots, p_{Nt}) + \sum_{i=1}^{N} (p_{it} - c) g_i(r_{it} - p_{it}) + \beta V(r_{1t+1}, r_{2t+1}, \ldots, r_{Nt+1}), \quad (13)$$

where $g_i = \delta_i$ if $r_{it} > p_{it}$, else $g_i = \gamma_i$; $\delta_i, \gamma_i > 0$; $r_{it} = \alpha r_{it-1} + (1-\alpha) p_{it-1}$, r_{i0} given.

We consider the following demand function:

$$d_{it} = a_i - b_i p_{it} + \sum_{j=1, j \neq i}^{N} \theta_{ij} p_{jt} + g_i(r_{it} - p_{it}), \quad (14)$$

where $g_i = \delta_i$ if $r_{it} \leq p_{it}$, else $g_i = \gamma_i$; r_{i0} given, $t = 1, 2, \ldots$, $\sum_{j=1, j \neq i}^{N} \theta_{ij} < b_i$, $i = 1, \ldots, N$.

When $\delta_i > \gamma_i$ for $i = 1, \ldots, N$, we prove that cyclical pricing is optimal and that the cycles are in phase. Further, if the brands are symmetric and the cycles are hi-lo, then the hi prices of all the brands are equal, as are the lo prices. These results are contained in Lemma A1 and Proposition A1 in the appendix. Thus, the behavior of the retailer is similar to that of a monopolist facing a demand function which is the sum of the individual brand functions. In the symmetric case, this reduces to

$$d_t = N[a - p_t(b - \theta(N-1)) + g(r_t - p_t)]$$

where, $g = \delta$ if $r_t \leq p_t$, else $g = \gamma$ and $(N-1)\theta < b$. The situation thus reverts to the case of a monopolist managing a single brand, with the corresponding results applying. When the brands are not symmetric, i.e., when the corresponding parameter values are different across brands, hi-lo prices *in phase* minimize the cross-price effect, while at the same time they allow the retailer to take advantage of the reference price effect.

We assess the optimal policy numerically by considering two brands and the demand function in Equation

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Table 1 Simulation Parameters, $\delta > \gamma$

$c = 0.5$, $a_i = 1$, $b_i = 1$, $\theta_{ij} = \theta = 0.2$, Time Horizon $T = 50$ Periods
Parameter values are selected such that the total demand does not exceed the total market potential, $a_1 + a_2$.

α	β	γ	δ
(0.02, 0 4, 0 6, 0.8)	(0 99, 1)	(0.1, 0.5)	(1.5γ, 2.0γ, 2.5γ)

(14). Note that the state variables are the reference prices of the two brands. Parameter values are shown in Table 1, and the study uses a full-factorial design. Since our focus is on the reference price effect, the cross-price effect is held constant at 0.2.

Our main findings are that the optimal pricing policy is hi-lo in phase and that for a given value of γ, the hi-lo range increases with δ. Hi-lo pricing in phase may seem counter intuitive, but in fact, it is frequently observed. Consider, as an example, the pricing strategies that supermarket chains use for Coke and Pepsi, two approximately symmetric brands. In some markets the local bottlers constrain the chains to not promote these brands during the same weeks, and consequently Coke and Pepsi are promoted in alternate weeks.[4] However, in other markets no such constraints are enforced. In these markets, many chains elect to promote Coke and Pepsi simultaneously.[5] When $\delta_i \leq \gamma_i$, our analytical results are more restricted, as in the monopoly case. We prove that constant prices are better than hi-lo pricing (see Proposition A2 in the appendix). We explore this issue numerically and determine the optimal pricing policy for a two-brand case. The optimization would identify more complex cycles if these are indeed superior to constant pricing. We consider two brands with the interdependent linear demand function given in Equation (14). Parameter values used in the full-factorial design are given in Table 2.

Our main finding is that the optimal pricing policy is constant. If the brands are symmetric, their prices are identical. This finding extends the single-brand case (Result 2) to include multiple brands, and the intuition underlying it is the same. When the loss parameters for the two brands are greater than the respective gain parameters, the total demand during a hi-lo pricing cycle will be lower than that would be obtained for a constant price. This result shows that a constant policy is optimal and thus generalizes Proposition A2.

To complete this section, we consider the case where there are N brands managed by the retailer, with n_1 of them viewed as having $\delta_i > \gamma_i$ and the remaining $N - n_1$ having $\delta_i \leq \gamma_i$. We prove in Proposition A3 (in the appendix) that it is not optimal to charge a constant price for any of the n_1 brands, and that it is more profitable to maintain constant prices than hi-lo prices for the remainder. We further explore this case numerically, considering two brands where for brand 1, $\delta_1 > \gamma_1$, and for brand 2, $\delta_2 \leq \gamma_2$. The parameter values used in a full-factorial design are shown in Tables 1 ($\delta_1 > \gamma_1$) and 2 ($\delta_2 \leq \gamma_2$). We find that the retailer uses a cyclical policy for brand 1 and a constant policy for brand 2. The intuition underlying this finding is as follows: in the exponential smoothing framework for the formation of reference prices, each brand has its own reference price, based only on its own past prices; the interdependency between the brands is only through the cross-price effect and not through the reference price effect. Thus, the impact on total sales of the reference price of each brand is independent, leading to independent pricing policies.

3.3. Duopoly

We now consider a duopoly—a scenario that has been analyzed in the price promotion literature referenced earlier. The difference, of course, is that we include reference price effects in the demand specification. As in §3.2 we assume that demand for the two brands 1 and

Table 2 Simulation Parameters, $\delta \leq \gamma$

$c = 0.5$, $a_i = 1$, $b_i = 0.2$, $\theta_{ij} = \theta = 0.2$, Time Horizon $T = 50$ Periods
Parameter values are selected such that the total demand does not exceed the total market potential, $a_1 + a_2$.

α	β	δ	γ
(0, 0.2, 0.4, 0.6, 0.8)	(0.99, 1)	(0.1, 0.5)	(1.0δ, 1.5δ, 2.0δ, 2.5δ)

[4] Examples are Abco and Safeway in Tucson, AZ.

[5] Examples are Star Markets in Boston, MA and Super Fresh stores in Baltimore, MD.

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2 is specified by Equation (14). The objective function for brand i ($i = 1, 2$) is given by:

$$\max_{p_i} \sum_{t=0}^{\infty} \beta^t (p_{it} - c)(a_i - b_i p_{it} + \theta_{ij} p_{jt} + g_i(r_{it} - p_{it})),$$

where $g_i = \delta_i$ if $r_{it} \leq p_{it}$, else $g_i = \gamma_i$, $\delta_i; \gamma_i > 0$; $r_{i,t} = \alpha r_{i,t-1} + (1 - \alpha) p_{i,t-1}$, r_{i0} given, $\theta_{ij} < b_i$.

We develop Markov equilibrium strategies in the class of games of perfect information with separable payoffs, i.e., Markov Perfect Equilibria (MPE) which yield a Nash equilibrium in every proper subgame (Fudenberg and Tirole 1992). First, we prove Lemma 1, which characterizes the equilibrium strategy in terms of whether the two firms should have cyclical prices *in phase* or *out of phase*. We then use this result to examine the relative impact of gain and loss parameters on the equilibrium solution.

For a homogeneous market where customers value gains more than losses for both brands, Proposition 3 characterizes the duopolists' optimal pricing strategy as cyclical. We explore the pricing policy using simulation and find that in all cases, the policy is hi-lo. Result 3 summarizes our findings. Next, we explore the case where for both brands, customers are loss averse. We prove in Proposition 4 that a constant price is superior to a hi-lo policy and further explore this finding numerically and conclude in Result 4 that a constant price is indeed optimal.

Finally, in Proposition 5, we examine the case where customers value gains more than losses for one brand ($\delta_1 > \gamma_1$) and are loss averse with regard to the other brand ($\delta_2 \leq \gamma_2$). This case approximates the tiered brand scenario discussed by Blattberg and Wisniewski (1989), where price cuts by lower tier brands do not substantially increase volume, whereas price cuts by upper tier brands do substantially increase volume. We show that the prices of brand 1 should cycle while brand 2 should maintain a constant price. Again, this idea is explored numerically, and in Result 5 we show that the pricing cycle for brand 1 is hi-lo. Finally, in Propositions A4, A5, and A6 (in the appendix), we extend the duopoly results to an oligopoly.

LEMMA 1. *Let demand be specified by Equation* (14) *for brand i, $i = 1, 2$. Then, under an average profit per period criterion, if the brand prices follow a two period cycle, the MPE Nash solution is to maintain the cyclical prices in phase.*

Lemma 1 is similar to Lemma A1 for the monopolist retailer and shows that if each player decides to maintain cyclical prices, cycling in phase is an equilibrium strategy and neither firm has an incentive to move to an *out of phase* cycle. The independent effect of each brand's reference price is driving this result. In §3.4 we find that when a dependency is introduced between the reference effects of the two brands, the duopolists face a prisoner's dilemma and will have a unilateral incentive to offer two consecutive lo prices and break the cycle. Below, we use Lemma 1 to show that maintaining a constant price is not an equilibrium strategy when gains are valued greater than losses.

PROPOSITION 3. i. *For $0 \leq \alpha < 1$, $\delta_i > \gamma_i$, $i = 1, 2$, demand specified by Equation* (14), *and an average profit per period criterion, the MPE Nash solution is not to have a constant price. Hence, the equilibrium policy cycles.*

ii. *Further, using an average profit per period criterion, if a hi-lo cycle is optimal and if the two firms are symmetric, the hi prices of the two brands are equal, as are the lo prices.*

When $\delta > \gamma$ it is in the best interest of the duopolists to take advantage of the net increase in total demand due to market expansion during any two period cycle. No firm has a unilateral incentive to deviate from a cyclical policy and offer, for example, lo prices in two consecutive periods since it will not be able to activate the reference price effects in future periods. Below, we characterize the optimal policy numerically. The state variables are the reference prices of the two brands. The parameter values used in the full-factorial simulation design are shown in Table 1. As in the monopolist retailer case, we consider a time horizon of 50 periods and analyze the pricing policy in periods 10 through 40 to remove any transient effects. The dynamic programming algorithm is solved using backward induction and interpolation where future reference prices are truncated at the second decimal place. During each stage of the game, we solve for the best response of each player given the other player's action and the resulting state. We searched for MPE Nash solutions and found unique solutions in all cases. The following result summarizes our findings.

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RESULT 3. For $0 \leq \alpha < 1$, $\delta > \gamma$, and demand specified by Equation (14), the MPE Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is characterized by hi-lo prices in phase for the two brands. Further, given γ, the hi-lo range increases with δ.

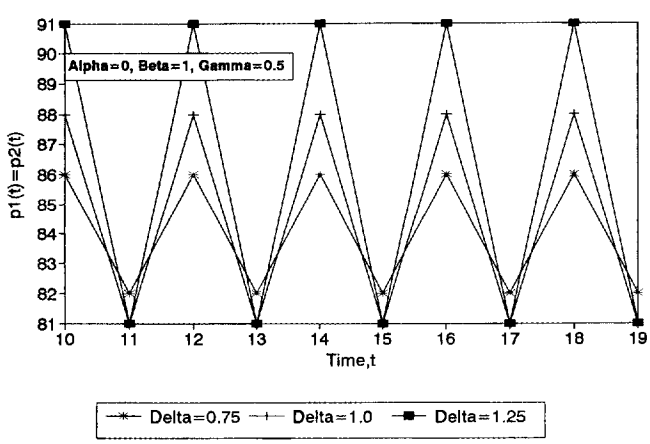
This result, shown in Figure 3 for $\alpha = 0$, $\beta = 1$, $\gamma = 0.5$, is similar to that obtained for a monopolist retailer managing two brands. The main difference is that the hi and lo prices are both higher in the case of a monopolist retailer than in the duopoly. Obviously, the total firm profits are lower in the duopoly case than in the monopolist retailer case; the latter corresponds to a joint maximum solution concept. We obtain similar results for the other levels of the parameters used in the simulation.

PROPOSITION 4. i. *For* $0 \leq \alpha < 1$, $\delta_i \leq \gamma_i$, $i = 1, 2$, *demand specified by Equation* (14), *and an average profit per period criterion, between the option of either charging a constant price or hi-lo prices, the MPE Nash solution is to maintain a constant price.*

ii. *Further, using an average profit per period criterion, if the two brands are symmetric, the constant prices of the brands are equal.*

Proposition 4 extends the monopoly result when gains are valued less than losses, to the case of a duopoly. Again, in order not to restrict ourselves to hi-lo policies and admit the possibility of more complex cycles, we characterize the optimal policy numerically. We use the full-factorial design shown in Table 2. The following result summarizes our findings:

Figure 3 **Symmetric Duopoly: Price Paths**



RESULT 4. For $0 \leq \alpha < 1$, $\delta \leq \gamma$, and demand specified by Equation (14), the *MPE* Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is characterized by a constant price.

Finally, we consider a case where customers value gains more than losses for brand 1 ($\delta_1 > \gamma_1$) and are loss-averse toward brand 2 ($\delta_2 \leq \gamma_2$). We have the following result.

PROPOSITION 5. *For* $0 \leq \alpha < 1$, $\delta_1 > \gamma_1$, $\delta_2 \leq \gamma_2$, *demand specified by Equation* (14), *and an average profit per period criterion, the MPE Nash solution is not to charge a constant price for brand* 1, *and between the option of charging a constant price or maintaining hi-lo prices, the MPE Nash solution is to maintain a constant price for brand* 2.

To further explore this case, we analyze this case numerically using the design outlined in Tables 1 and 2. The following result summarizes our findings.

RESULT 5. For $0 \leq \alpha < 1$, demand specified by Equation (14), and two brands with $\delta_1 > \gamma_1$ and $\delta_2 \leq \gamma_2$, the MPE Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is characterized by a hi-lo policy for brand 1 and a constant price for brand 2. Further, given γ_1, the hi-lo range for brand 1 increases with δ_1.

The duopolists face a market where the interdependence between the brands is captured only by the cross-price effect; the reference price effects operate independently of one another. Therefore, when gains are greater than losses for only one brand, the increase in the demand for that brand during any pricing cycle is primarily from market expansion. Thus, the firm reaps higher profits without being in any danger of facing a threat from its competitor.

Propositions A4, A5, and A6 in the appendix extend the duopoly results to an oligopoly with N brands. Table 3 summarizes the results obtained in this section.

3.4. Pricing Strategies Under Reference Brand Formulation

Are the findings in the previous sections reflective of the genuine importance of the gain/loss parameters, or

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Table 3 Summary of Results: Exponential Smoothing Formulation for Reference Price Formation

Framework	$\delta_i > \gamma_i$	$\delta_i \leq \gamma_i$	$\delta_i > \gamma_i, 1 \leq i \leq n_1$	$\delta_j \leq \gamma_j, 1 \leq j \leq N - n_1$
Monopoly	Hi-lo	Constant	Not Applicable	Not Applicable
Monopolist Retailer	Hi-lo (in phase)	Constant	Hi-lo	Constant
Duopoly	Hi-lo (in phase)	Constant	Hi-lo	Constant
Oligopoly	Hi-lo (in phase)	Constant	Hi-lo	Constant

are they a consequence of the specific reference price formation process we have used? In this section we investigate the robustness of our results by using the reference brand formulation proposed by Hardie et al. (1993) and discussed in §2.1. We first develop an aggregate version of the Hardie et al. formulation and then explore policy implications numerically.

Consider a market with two brands 1 and 2 and assume that a consumer has bought at least one brand in the past. At the beginning of any period t, a fraction of customers Θ_{t-1}, has bought brand 1 in their last purchase, and the remaining fraction $(1 - \Theta_{t-1})$ has bought brand 2 in their last purchase. Of the fraction Θ_{t-1}, some will buy brand 1, some will buy brand 2, and the remainder will buy nothing in period t. Let $d_{1t|1}$ and $d_{2t|1}$ represent the corresponding demands for brands 1 and 2, respectively. Assuming a linear demand function, and using the reference brand formulation, we have:

$$d_{1t|1} = a_1 - b_1 p_{1t} + \theta_{12} p_{2t},$$

$$d_{2t|1} = a_2 - b_2 p_{2t} + \theta_{21} p_{1t} + g_2(p_{1t} - p_{2t}),$$

where, $g_2 = \delta_2$ if $p_{1t} > p_{2t}$, else $g_2 = \gamma_2$, $\theta_{12} < b_1$, and $\theta_{21} < b_2$.

Similarly, considering the fraction $(1 - \Theta_{t-1})$, we have:

$$d_{2t|2} = a_2 - b_2 p_{2t} + \theta_{21} p_{1t},$$

$$d_{1t|2} = a_1 - b_1 p_{1t} + \theta_{12} p_{2t} + g_1(p_{2t} - p_{1t}),$$

where, $g_1 = \delta_1$ if $p_{2t} > p_{1t}$, else $g_1 = \gamma_1$.

Therefore, td_{1t}, the total demand for brand 1 in period t is given by, $\Theta_{t-1} d_{1t|1} + (1 - \Theta_{t-1}) d_{1t|2}$. Similarly, $td_{2t} = \Theta_{t-1} d_{2t|1} + (1 - \Theta_{t-1}) d_{2t|2}$. The parameters δ_i and γ_i, $i = 1, 2$, are chosen such that the total demand for the two brands $td_{1t} + td_{2t}$ cannot exceed the market potential, $a_1 + a_2$ (obtained when all prices are set to zero). Thus, the proportion of market that bought brand 1 in period t is given by

$$\Theta_{t-1} \frac{d_{1t|1}}{(a_1 + a_2)} + (1 - \Theta_{t-1}) \frac{d_{1t|2}}{(a_1 + a_2)}.$$

Thus, at the beginning of period $t + 1$, the fraction of customers who bought brand 1 in their last purchase,

$$\Theta_t = \Theta_{t-1} \frac{d_{1t|1}}{(a_1 + a_2)} + (1 - \Theta_{t-1}) \frac{d_{1t|2}}{(a_1 + a_2)} + \Theta_{t-1}\left(1 - \frac{d_{1t|1} + d_{2t|1}}{(a_1 + a_2)}\right).$$

We analyze this problem in a two-brand monopoly as well as in a duopoly. Given the complexity of this specification, we could not obtain analytical results; therefore, our results are based on simulation.

Monopolist Retailer Managing Two Brands

The objective function in this case is:

$$\max_{p_1, p_2} \sum_{t=0}^{\infty} \beta^t((p_{1t} - c)td_{1t} + (p_{2t} - c)td_{2t}). \quad (15)$$

Accordingly, we have the following recursive relation:

$$H(V)(\Theta_t, p_{1t}, p_{2t}) = (p_{1t} - c)td_{1t} + (p_{2t} - c)td_{2t} + \beta V(\Theta_{t+1}). \quad (16)$$

We vary the model parameters using the full-factorial design shown in Tables 1 ($\delta > \gamma$) and 2 ($\delta \leq \gamma$), and discussed in §3.2. Since the focus is on the effect of reference prices, the cross-price effect θ is kept constant at 0.2. The following results summarize our findings:

RESULT 6. For $\delta > \gamma$, the optimal pricing policy that solves (15) over the time horizon considered either (a) with discounting or (b) using the average profit per period criterion, is characterized by hi-lo prices for both

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the brands and out of phase. Further, given γ, the hi-lo range increases with δ.

When gains are valued greater than losses, the total demand during a two-period cycle increases when brands have cyclical prices. Under the reference brand formulation, if the monopolist retailer maintains hi-lo prices for the two brands *in phase*, the total demand for the two brands is not increased since the reference brand effect is not activated. For example, in period $t - 1$, let the two brands have equal prices (i.e., both hi or both lo). In period t, for those who bought brand i in period $t - 1$, we have:

$$d_{it|i} = a_i - b_i p_{it} + \theta_{ij} p_{jt}, \quad d_{jt|i} = a_j - b_j p_{jt} + \theta_{ji} p_{it}.$$

The retailer can take advantage of the process only if the prices of the two brands cycle *out of phase*. Under the exponential smoothing formulation, an increase in the current price of a brand yields a future benefit to the same brand; whereas, in the reference brand formulation an increase in the current price of a brand will only increase the demand for the other brand. Thus, the phase difference in the pricing cycle of the two brands is the main distinction between the effects of reference brand formulation and the exponential smoothing process on optimal prices.

RESULT 7. For $\delta \leq \gamma$, the optimal pricing policy that solves (15) over the time horizon considered either (a) with discounting or (b) using the average profit per period criterion, is a constant price for both the brands.

The above result parallels the monopolist retailer case discussed in §3.2. When $\delta \leq \gamma$ the absolute change in demand during any pricing cycle for the two brands is negative; therefore, the monopolist retailer does not have an incentive to maintain cyclical prices. The intuition for this result is the same as in the previous scenarios, i.e., since the losses are valued greater than the gains, the total demand during a hi-lo pricing cycle will be lower than the total demand obtained by maintaining a constant price. Hence, the monopolist retailer does not have an incentive to have a cyclical pricing policy. Below, we consider the case where customers value gains more than losses for brand 1 ($\delta_1 > \gamma_1$) and are loss averse toward brand 2 ($\delta_2 \leq \gamma_2$).

RESULT 8. For $\delta_1 > \gamma_1$, and $\delta_2 \leq \gamma_2$, the optimal pricing policy that solves (15) over the time horizon considered either (a) with discounting or (b) using the average profit per period criterion, is a constant price for both the brands.

This result is in contrast to the corresponding finding for the exponential smoothing process. Under the reference brand formulation, the absolute change in total demand during any pricing cycle will be positive only when gains are valued greater than losses for both the brands and when the prices are out of phase. If the brands are asymmetric, the net change in demand using a cyclical policy is not positive. The positive impact on one brand is nullified by the negative impact on the other. Thus, the monopolist retailer does not have an incentive to maintain cyclical prices for any brand.

Duopoly

The objective function for firm i in this case is:

$$\max_{p_i} \sum_{t=0}^{\infty} \beta^t (p_{it} - c) t d_{it}, \quad i = 1, 2. \tag{17}$$

Accordingly, we have the following recursive relation for firm i:

$$H(V)(\Theta_t, p_{it}, p_{jt}) = (p_{it} - c) t d_{it} + \beta V(\Theta_{t+1}), \quad i = 1, 2. \tag{18}$$

As in the duopoly case using the exponential smoothing process (§3.3), we consider a time horizon of 50 periods and analyze the pricing policy in periods 10 through 40 to remove any transient effects. The dynamic programming algorithm is solved using backward induction and interpolation where future reference prices are truncated at the second decimal place. During each stage of the game, we solve for the best reply of each player given the other player's action and the resulting state. We searched for MPE Nash solutions and found unique solutions in all cases.

We vary the model parameters using the full-factorial design shown in Tables 1 ($\delta > \gamma$) and 2 ($\delta \leq \gamma$). Since the focus is on the effect of reference prices, the cross-price effect θ, is kept constant at 0.2. The optimal solution is obtained numerically and the following results summarize our findings.

RESULT 9. For $\delta > \gamma$, and the objective function specified by (17), the MPE Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is a constant price for both the brands.

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Table 4 Summary of Results: Reference Brand Formulation for Reference Price Formation

Framework	$\delta_i > \gamma_i, i = 1, 2$	$\delta_i \leq \gamma_i, i = 1, 2$	$\delta_1 > \gamma_1, \delta_2 \leq \gamma_2$
Monopoly	Not Applicable	Not Applicable	Not Applicable
Monopolist Retailer	Hi-lo (out of phase)	Constant	Constant
Duopoly	Constant	Constant	Constant

As seen in the monopolist retailer case above, the joint profit maximizing strategy for the duopolists is to maintain cyclical prices *out of phase*. But in a duopoly, a prisoner's dilemma exists due to the interdependence between the two brands created by the reference brand formulation. This interdependence provides a unilateral incentive for each firm to defect and not lose the consumers who switch to the cheaper brand during the hi price period. This is achieved by offering, say, two consecutive low prices and thus breaking the out of phase sequence. Unless a punishment strategy is enforced by the two firms, joint profit maximization with pricing cycles out of phase cannot be an equilibrium strategy under the reference brand formulation.

RESULT 10. For $\delta \leq \gamma$, and the objective function specified by (17), the MPE Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is to maintain a constant price for both the brands.

When $\delta \leq \gamma$ the total demand during a pricing cycle is lower than if the prices are constant. This is true in both the exponential smoothing process and the reference brand formulation. Below, we consider the case where customers value gains more than losses for brand 1 ($\delta_1 > \gamma_1$) and are loss averse toward brand 2 ($\delta_2 \leq \gamma_2$).

RESULT 11. For $\delta_1 > \gamma_1$ and $\delta_2 \leq \gamma_2$, and the objective function specified by (17), the MPE Nash solution that maximizes either (a) the sum of discounted profits over the time horizon considered or (b) the average profit per period, is a constant price for both brands.

The above result highlights another major implication of the reference price formation process. Under the reference brand formulation, each firm's pricing decisions affect the other firm in two ways, one through the cross-price effect and the other via the reference price effect. Therefore, neither brand can increase its demand by maintaining cyclical prices. This does not happen in the exponential smoothing process because the selection of a brand in any period is independent of prior choices (Table 4).

3.5. Impact of Heterogeneity

We now relax the assumption of homogeneous customers who view each brand in the same way, either valuing gains more than losses or the opposite. We assume that there are two segments in the market: Segment 1, which values gains more than losses; and Segment 2, which is loss averse. We want to determine if there is some critical proportion of Segment 1 customers which will yield an optimal policy which is cyclical. We first study the case where the reference price of a brand is the exponentially smoothed value of past brand prices. Suppose Segment 1 and Segment 2 have memory parameters α_1 and α_2, gain parameters δ_1 and δ_2, and loss parameters γ_1 and γ_2, such that $\delta_1 > \gamma_1$ and $\delta_2 \leq \gamma_2$. Let q_1 and $(1 - q_1)$ be the shares of the two segments. We examine the monopoly, monopolist retailer managing symmetric brands, symmetric duopoly and oligopoly cases. For all these cases we obtain the same condition for the proportion of Segment 1. The condition is stated in Proposition 6.

PROPOSITION 6. *Using an exponential smoothing process of reference price formation, for Segments* 1 *and* 2, *where* $\delta_1 > \gamma_1, \delta_2 \leq \gamma_2$, *demand specified by Equation* (14), *and under an average profit per period criterion, a sufficient condition for a constant pricing policy not to be optimal in a single brand monopoly, a symmetric multibrand monopoly, a symmetric duopoly, or a symmetric oligopoly, is*

$$\left(\frac{q_1}{1 - q_1}\right) > \left(\frac{1 + \alpha_1}{1 + \alpha_2}\right)\left(\frac{\gamma_2 - \delta_2}{\delta_1 - \gamma_1}\right).$$

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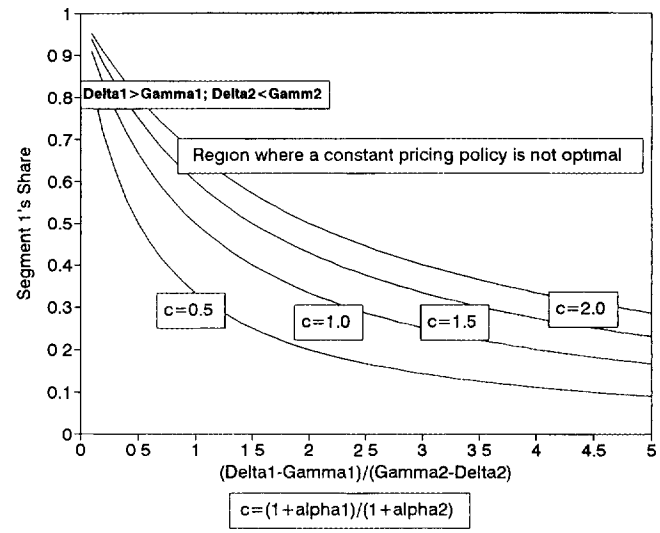
Figure 4 plots the critical value of q_1 as a function of the other parameters; when q_1 exceeds the critical value, constant prices are not optimal.

We set $(1 + \alpha_1)/(1 + \alpha_2) = c$. When Segment 1 has a one-period memory ($\alpha_1 = 0$), and Segment 2 has a much longer memory, c tends to 0.5, while when the memory assumptions are reversed, c tends to 2.0. Note that as c decreases, the critical value of q_1 (the share of Segment 1), also decreases. Therefore, if Segment 1 has a very short memory, cyclical policies are optimal for relatively small values of q_1. For example, when $(\delta_1 - \gamma_1)/(\gamma_2 - \delta_2)$ equals 1, q_1 must exceed 0.33 for cyclical policies to be optimal and this value decreases as $(\delta_1 - \gamma_1)/(\gamma_2 - \delta_2)$ increases. When $(\delta_1 - \gamma_1)/(\gamma_2 - \delta_2)$ is 3, the critical value of q_1 is only 0.15. Thus, if Segment 1 has a short memory compared to Segment 2 and the reference price effect is greater in Segment 1 than in Segment 2, a small value of q_1 will lead to a cyclical policy. If the memory parameters are equal, and $((\delta_1 - \gamma_1)/(\gamma_2 - \delta_2))$ is 2, then q_1 must exceed 0.33, which reduces to 0.25 when $((\delta_1 - \gamma_1)/(\gamma_2 - \delta_2))$ increases to 3. Note that these are sufficient conditions for the critical levels of q_1, i.e., they represent the most stringent requirements. If we follow Krishnamurthi et al. (1992) and assume Segment 1 are switchers while Segment 2 are loyals, then an empirical question is: "How likely is it that switchers account for at least 33% (for example) of the market?"[6]

For the reference brand formulation we explored the critical value of q_1 using numerical methods. Our findings are consistent with those reported for the exponential smoothing formulation, namely, a relatively small value of q_1 leads to cyclical policies in all cases where cyclical policies are optimal in a homogeneous market consisting of Segment 1 buyers only.

Figure 4 Impact of Two Segments (1 and 2) on Pricing Policy



4. Discussion

In this paper we have shown that when demand is a function of reference prices, cyclical prices may be optimal. We considered two reference price formation processes. In the first, the reference price formation is characterized as an adaptive expectations process and operationalized as the exponentially smoothed (ES) value of past prices. The impact of reference price on demand is then expressed as the difference between the reference price (r_t) and the observed price (p_t). When the difference is positive, customers perceive a gain and demand increases by an amount $\delta(r_t - p_t)$. When the difference is negative, customers perceive a loss and demand decreases by an amount $\gamma(p_t - r_t)$. In this formulation, if there are multiple brands, the reference price of each brand is assumed to be a function of its own past prices. Interaction between the brands occurs only via the observed prices, i.e., cross-price effect. The second reference price formation process we considered assumes that the reference price is the current price of the brand last purchased, the reference brand (RB). The price of any brand being considered for purchase is compared to the price of the reference brand; the impact of reference price on demand is similar to that in the ES

[6] For two categories where published data are available, this percentage appears to be quite reasonable Kannan and Sanchez (1994) define switchers as those who have switched at least once during the modeling period, and report that for the coffee market consisting of 11 brands, switchers account for at least 59% of each brand's sales, and account for about 72% of total market. Similarly, they report for the cracker market (13 brands) that switchers account for at least 82% of each brand's sales, and account for about 92% of total market. Clearly, these numbers are upper bounds on the sizes of the respective switching segments if we use Krishnamurthi et al.'s framework. However, our requirements for cyclical policies would be met even if less than one half of these segments satisfied criterion of Krishnamurthi et al.

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formulation. Consequently, the interaction between brands occurs not only via the cross-price effect but also through reference prices. In terms of the pricing strategy, the most important difference between the ES and the RB formulations can be summarized as follows: when $\delta > \gamma$, in the ES case, an increase in the current price of a brand leads to a future benefit to that brand, because by increasing the future reference price it reduces the impact of a price increase, and enhances the impact of a price decrease. In the RB case, a similar price increase leads to an increase in the sales of the competing brand. It has no beneficial effect in the future on the brand under consideration unless the competing brand reciprocates.

For the ES formulation we developed the optimal strategy in a monopoly, for a monopolist retailer managing multiple brands, a duopoly, and an oligopoly. For the RB formulation, the monopoly scenario is infeasible, and we considered a monopolist retailer managing two brands and a duopoly. Cyclical pricing policies were obtained only when the gain parameter (δ) was greater than the loss parameter (γ). In the ES formulation, when $\delta > \gamma$, the pricing policy is cyclical and in phase for a monopolist retailer as well as for a duopolist and an oligopolist. In contrast, when $\delta > \gamma$ in the RB formulation, the pricing policy is cyclical and out of phase for the monopolist retailer, and a constant policy for the duopolist. Out of phase cyclical prices in a duopoly lead to a prisoner's dilemma and the non-cooperative solution is a constant price. On the other hand, the monopolist retailer maximizes total profits from the two brands and is therefore able to maintain *out of phase* cyclical prices for the two brands. Under the ES formulation, there is no prisoner's dilemma in a duopoly since there is no incentive for one of the duopolists to offer two successive lo prices. In both formulations, when $\delta \leq \gamma$ constant prices are optimal.

Examining retailer practice, we find that a majority of grocery supermarket chains follow hi-lo policies, although there is interest in everyday low price (EDLP) strategies (Partch 1992). However, when hi-lo policies are followed, they are usually not followed for all brands in a category but primarily for the leading brands. As noted earlier, both simultaneous promotions and alternating promotions of symmetric brands are observed. In field experiments, Hoch et al. (1994) find that hi-lo policies far outperform EDLP policies. They consider 19 product categories and find that hi-lo prices led to a 15% increase in profitability over control stores, while EDLP led to a 18% decrease in profits. They also find that while demand is much greater when prices are low, there is substantial demand even when prices are hi. Clearly, not all customers program their buying to anticipate the low price periods and make all purchases at that time. The Hoch et al. (1994) findings suggest that as more manufacturers move away from trade dealing and to constant pricing to retailers, price promotions to consumers will still continue to be a prominent retailing strategy. These observations provide some anecdotal support for our analytical conclusions.

Appendix

Monopoly

PROOF OF PROPOSITION 1. The proof is by contradiction. Assume there is a r such that $p(r) = r$. Let $r_0 = r$ be the reference price in an arbitrarily chosen period 0 and let p_t be the price in period t. Thus, $r_t = \alpha^t r + (1-\alpha)\sum_{j=0}^{t-1} \alpha^{t-1-j} p_j$. Since $p(r) = r$, we have $V(r) = f(r)/(1-\beta)$, which must be larger than always charging $(r+\Delta)$ or $(r-\Delta)$, $\Delta > 0$. Hence,

$$\frac{f(r)}{1-\beta} \geq \frac{f(r+\Delta)}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \gamma (r+\Delta-c)(r_t - (r+\Delta)),$$

$$\frac{f(r)}{1-\beta} \geq \frac{f(r-\Delta)}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \delta (r-\Delta-c)(r_t - (r-\Delta)).$$

But,

$$p_t = r+\Delta \Rightarrow r_t = \alpha^t r + (1-\alpha)\frac{1-\alpha^t}{1-\alpha}(r+\Delta) = r + (1-\alpha^t)\Delta.$$

Similarly, $p_t = r - \Delta \Rightarrow r_t = r - (1-\alpha^t)\Delta$. Hence,

$$r_t - (r+\Delta) = r + \Delta(1-\alpha^t) - r - \Delta = -\alpha^t\Delta$$

and

$$r_t - (r-\Delta) = r - \Delta(1-\alpha^t) - r + \Delta = \alpha^t\Delta.$$

Thus, we have,

$$\frac{f(r)}{1-\beta} \geq \frac{f(r+\Delta)}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \gamma (r+\Delta-c)(-\alpha^t\Delta)$$

$$= \frac{f(r+\Delta)}{1-\beta} - \frac{\gamma\Delta}{1-\alpha\beta}(r+\Delta-c).$$

Similarly,

$$\frac{f(r)}{1-\beta} \geq \frac{f(r-\Delta)}{1-\beta} + \frac{\delta\Delta}{1-\alpha\beta}(r-\Delta-c),$$

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which can be rewritten as:

$$\frac{f(r+\Delta)-f(r)}{1-\beta} \leq \frac{\gamma\Delta}{1-\alpha\beta}(r+\Delta-c)$$

$$\frac{f(r)-f(r-\Delta)}{1-\beta} \geq \frac{\delta\Delta}{1-\alpha\beta}(r-\Delta-c).$$

If f is continuous and differentiable, let $f'(r)$ denote its derivative with respect to r. Taking the limit as $\Delta \to 0$, and combining above two equations, we have,

$$\frac{\delta}{1-\alpha\beta}(r-c) \leq \frac{f'(r)}{1-\beta} \leq \frac{\gamma}{1-\alpha\beta}(r-c),$$

which is impossible if $\delta > \gamma$. If f is convex,

$$\frac{\delta}{1-\alpha\beta}(r-c) \leq \frac{f(r)-f(r-\Delta)}{\Delta(1-\beta)}$$

$$\leq \frac{f(r+\Delta)-f(r)}{\Delta(1-\beta)} \leq \frac{\gamma}{1-\alpha\beta}(r-c),$$

which is impossible if $\delta > \gamma$. Thus, $\exists r$: $p(r) = r$. □

PROOF OF PROPOSITION 2.i. Let $p_0, p_1, p_2, \ldots, p_n$ be the optimal sequence of prices in periods $0, 1, \ldots, n$, when the initial reference price is r_0. Thus,

$$V(r_0) = \sum_{t=0}^{n} \beta^t(p_t(r_0)-c)\delta(r_t(r_0)-p_t(r_0)).$$

For $\Delta > 0$, we have,

$$V(r_0+\Delta) = \sum_{t=0}^{n} \beta^t(p_t(r_0+\Delta)-c)\delta(r_t(r_0+\Delta)-p_t(r_0+\Delta)).$$

Noting that $p_t(r_0+\Delta)$ need not equal $p_t(r_0)$, we write:

$$V(r_0+\Delta) \geq \sum_{t=0}^{n} \beta^t(p_t(r_0)-c)\delta((r_t(r_0+\Delta)|_{p_t(r_0)})-p_t(r_0)).$$

If $p_t(r_0)$ is charged instead of $p_t(r_0+\Delta)$ we have, $r_t(r_0+\Delta)|_{p_t(r_0)} = r_t(r_0)|_{p_t(r_0)} + \alpha^t\Delta$. Thus,

$$V(r_0+\Delta) \geq V(r_0) + \sum_{t=0}^{n} \beta^t(p_t-c)\delta\alpha^t\Delta. \tag{19}$$

Similarly, since $r_t(r_0-\Delta)|_{p_t(r_0)} = r_t(r_0)|_{p_t(r_0)} - \alpha^t\Delta$, we have,

$$V(r_0-\Delta) \geq V(r_0) - \sum_{t=0}^{n} \beta^t(p_t-c)\delta\alpha^t\Delta. \tag{20}$$

Adding 19 and 20 we get,

$$(V(r_0+\Delta)-V(r_0))-(V(r_0)-V(r_0-\Delta)) \geq 0. \tag{21}$$

From 19 we see that V is nondecreasing in r; hence, 21 shows V is convex in r. Since p_0 and p_1 are the optimal prices in periods in 0 and 1, respectively, we have,

$$V(r_0) = f(p_0) + \delta(p_0-c)(r_0-p_0) + \beta V(r_1)$$

$$\geq f(p_1) + \delta(p_1-c)(r_0-p_1) + \beta V(r_1+\Delta'), \tag{22}$$

where $r_1 = \alpha r_0 + (1-\alpha)p_0$ and $\Delta' = (1-\alpha)(p_1-p_0)$. Note that Δ' can be positive, zero, or negative depending on whether p_1 is greater than, equal to or less than p_0. Similarly we have,

$$V(r_1) = f(p_1) + \delta(p_1-c)(r_1-p_1) + \beta V(r_2)$$

$$\geq f(p_0) + \delta(p_0-c)(r_1-p_0) + \beta V(r_2-\Delta'), \tag{23}$$

where $r_2 = \alpha r_1 + (1-\alpha)p_1$. Adding 22 and 23, we obtain upon simplification,

$$\beta\{(V(r_2)-V(r_2-\Delta'))-(V(r_1+\Delta')-V(r_1))\}$$

$$\geq \delta(p_0-p_1)(r_1-r_0). \tag{24}$$

By an inductive argument, the following is a sufficient condition to prove the monotonicity of the price path: if $p_0 < r_0$, then $p_1 \leq p_0$; if $p_0 > r_0$, then $p_1 \geq p_0$. We need to analyze a total of eight cases The four possible locations of p_1 given $p_0 < r_0$ are the following four segments s_1 through s_4:

$$c\overset{s_1}{\text{———}}p_0\overset{s_2}{\text{———}}r_1\overset{s_3}{\text{———}}r_0\overset{s_4}{\text{———}}.$$

The remaining four locations of p_1 when $p_0 > r_0$ are the segments s_5 through s_8:

$$c\overset{s_5}{\text{———}}r_0\overset{s_6}{\text{———}}r_1\overset{s_7}{\text{———}}p_0\overset{s_8}{\text{———}}$$

We will now show that when $p_0 < r_0$, p_1 will lie in segment s_1 and when $p_0 > r_0$, p_1 will lie in segment s_8. When $r_0 = p_0$, then $p_1 = p_0$ since the optimal response to r_0 is p_0.

Case 1 (p_1 *in* s_4). $p_0 \leq r_1 < r_0 \leq p_1$. Thus, substituting in (24) we have

$$(V(r_2)-V(r_2-\Delta'))-(V(r_1+\Delta')-V(r_1)) > 0.$$

Since V is increasing in r, this implies that V is strictly concave since $r_2-\Delta' < r_1 < r_2 < r_1+\Delta'$, and $\Delta' > 0$. This is a contradiction of 21.

Case 2 (p_1 *in* s_3). $p_0 \leq r_1 < p_1 < r_0$. Again, in this case we find that

$$(V(r_2)-V(r_2-\Delta'))-(V(r_1+\Delta')-V(r_1)) > 0,$$

which is a contradiction of 21 since $r_2-\Delta' < r_1 < r_2 < r_1+\Delta'$, $\Delta' > 0$, and V is nondecreasing in r.

Case 3 (p_1 *in* s_2). $p_0 < p_1 \leq r_1 < r_0$. Again,

$$(V(r_2)-V(r_2-\Delta'))-(V(r_1+\Delta')-V(r_1)) \geq 0,$$

which is a contradiction of 15 since $r_2-\Delta' < r_2 < r_1 < r_1+\Delta'$, $\Delta' > 0$, and V is nondecreasing in r

Case 4 (p_1 *in* s_1). $p_1 \leq p_0 < r_1 < r_0$. Here we find,

$$(V(r_2)-V(r_2-\Delta'))-(V(r_1+\Delta')-V(r_1))$$

$$\geq \text{nonpositive number},$$

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which is not a contradiction of the convexity of V since $r_2 < r_1 + \Delta' < r_2 - \Delta' < r_1$, $\Delta' \leq 0$, and V is nondecreasing in r.

Thus, when $\delta = \gamma$, if $p_0 < r_0$, then $p_1 \leq p_0$. Similarly, we can show that when $p_0 > r_0$, we do not arrive at a contradiction in the case where $p_1 \geq p_0 \geq r_1 > r_0$. Note that the sequence of prices p_t are a function of initial reference price r_0. Hence, the convergent price p^{**} is also a function of r_0.

PROOF OF PROPOSITION 2 ii Let p_0, p_1 be the optimal prices in periods 0 and 1, respectively, when the initial reference price is r_0. With a one-period memory, we have $r_t = p_{t-1}$. As noted earlier, the following is a sufficient condition to prove the monotonicity of the price path when $\gamma > \delta$: if $p_0 < r_0$, then $p_1 \leq p_0$; if $p_0 > r_0$, then $p_1 \geq p_0$. We need to analyze a total of six cases. The first three cases have $p_0 < p_0$. The other three cases are the mirror images.

Case 1 ($p_0 < r_0 \leq p_1$) Since p_0 is the optimal price in period 0 we have,

$$f(p_0) + (p_0 - c)\delta(r_0 - p_0) + \beta V(p_0) \geq f(p_1) + (p_1 - c)\gamma(r_0 - p_1) + \beta V(p_1),$$

which implies,

$$\beta(V(p_1) - V(p_0)) \leq f(p_0) - f(p_1) + (p_0 - c)\delta(r_0 - p_0) - (p_1 - c)\gamma(r_0 - p_1). \quad (25)$$

Similarly, since p_1 is the optimal price in period 1, we have,

$$V(p_0) = f(p_1) + (p_1 - c)\gamma(p_0 - p_1) + \beta V(p_1) \geq f(p_0) + (p_0 - c)\delta(p_0 - p_0) + \beta V(p_0),$$

which implies,

$$\beta(V(p_1) - V(p_0)) \geq f(p_0) - f(p_1) - (p_1 - c)\gamma(p_0 - p_1). \quad (26)$$

Subtracting (25) from (26) we have,

$$0 \geq -(p_1 - c)\gamma(p_0 - p_1) - (p_0 - c)\delta(r_0 - p_0) + (p_1 - c)\gamma(r_0 - p_1) = (r_0 - p_0)(\gamma(p_1 - c) - \delta(p_0 - c)),$$

which is positive ($p_0 \geq c$ by assumption and $p_1 > p_0$) and hence a contradiction.

Case 2 ($p_0 < p_1 < r_0$). Since p_0 is the optimal price in period 0 we have,

$$f(p_0) + (p_0 - c)\delta(r_0 - p_0) + \beta V(p_0) \geq f(p_1) + (p_1 - c)\delta(r_0 - p_1) + \beta V(p_1),$$

which implies,

$$\beta(V(p_1) - V(p_0)) \leq f(p_0) - f(p_1) + (p_0 - c)\delta(r_0 - p_0) - (p_1 - c)\delta(r_0 - p_1). \quad (27)$$

Equation (26) holds here Subtracting (27) from (26) we get,

$$0 \geq -(p_1 - c)\gamma(p_0 - p_1) - (p_0 - c)\delta(r_0 - p_0) + (p_1 - c)\delta(r_0 - p_1) \geq -(p_1 - c)\delta(p_0 - p_1) - (p_0 - c)\delta(r_0 - p_0) + (p_1 - c)\delta(r_0 - p_1)$$

since $\gamma > \delta = (r_0 - p_0)\delta(p_1 - p_0)$,

which is positive and hence a contradiction.

Case 3 ($p_1 \leq p_0 < r_0$). Since p_0 is the optimal price in period 0 we have,

$$f(p_0) + (p_0 - c)\delta(r_0 - p_0) + \beta V(p_0) \geq f(p_1) + (p_1 - c)\delta(r_0 - p_1) + \beta V(p_1),$$

which implies,

$$\beta(V(p_1) - V(p_0)) \leq f(p_0) - f(p_1) + (p_0 - c)\delta(r_0 - p_0) - (p_1 - c)\delta(r_0 - p_1). \quad (28)$$

Similarly, since p_1 is the optimal price in period 1, we have,

$$V(p_0) = f(p_1) + (p_1 - c)\delta(p_0 - p_1) + \beta V(p_1) \geq f(p_0) + (p_0 - c)\delta(p_0 - p_0) + \beta V(p_0),$$

which implies,

$$\beta(V(p_1) - V(p_0)) \geq f(p_0) - f(p_1) - (p_1 - c)\delta(p_0 - p_1). \quad (29)$$

Subtracting (28) from (29), we have,

$$0 \geq -(p_1 - c)\delta(p_0 - p_1) - (p_0 - c)\delta(r_0 - p_0) + (p_1 - c)\delta(r_0 - p_1) = (r_0 - p_0)\delta(p_1 - p_0),$$

which is nonpositive. Thus, we do not arrive at a contradiction when $p_1 \leq p_0 < r_0$. Proceeding as above, we find that in the remaining three cases, when $p_0 > r_0$, i.e., $p_1 \leq r_0 < p_0$, $r_0 < p_1 < p_0$, and $r_0 < p_0 \leq p_1$, we do not arrive at a contradiction in the last case. Hence, when $\delta < \gamma$, if $p_0 < r_0$, then $p_1 \leq p_0$; if $p_0 > r_0$, then $p_1 \geq p_0$.[7]

PROOF OF PROPOSITION 2.iii. Note that since we are considering finite steady state cycles, the criterion that we use is average profit per period. We will first consider a two period cycle and show that charging a single price which maximizes f is better than charging two different prices. Let p_0 and p_1 be the optimal prices in the two periods such that $p_0 < p_1$ Let the corresponding reference prices be r_0 and r_1. We have,

$$r_0 = \alpha r_1 + (1 - \alpha)p_1, \quad \text{and}$$

$$r_1 = \alpha r_0 + (1 - \alpha)p_0 = \alpha(\alpha r_1 + (1 - \alpha)p_1) + (1 - \alpha)p_0,$$

which implies, upon simplification,

[7] If $r_0 = p_0$, then $p_1 = p_0$ since the optimal response to r_0 is p_0. Also, the sequence of prices p_t are a function of initial reference price r_0. Hence, the convergent price p^{**} is also a function of r_0.

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$$r_1 = \frac{p_0 + \alpha p_1}{1 + \alpha} > \frac{p_0 + \alpha p_0}{1 + \alpha} = p_0.$$

Similarly, we have,

$$r_0 = \frac{p_1 + \alpha p_0}{1 + \alpha} < \frac{p_1 + \alpha p_1}{1 + \alpha} = p_1.$$

Thus we have the following inequality: $p_1 > r_0 > r_1 > p_0$. The average profit in each period during the cycle is given by,

$$\begin{aligned}\text{AP1} &= \{f(p_0) + (p_0 - c)\delta(r_0 - p_0) + f(p_1) + (p_1 - c)\gamma(r_1 - p_1)\}/2\\ &< \{f(p_0) + f(p_1) + \delta((p_0 - c)(r_0 - p_0) + (p_1 - c)(r_1 - p_1))\}/2\\ &< \{f(p_0) + f(p_1) + \delta((p_1 - c)(r_0 + r_1 - (p_0 + p_1)))\}/2\\ &= \{f(p_0) + f(p_1)\}/2.\end{aligned}$$

Let p^* maximize $f(p)$. By definition, $f(p^*) \geq f(p_0)$; and $f(p^*) \geq f(p_1)$. Now consider a constant pricing policy of p^*. Average profit in two periods is given by:

$$\text{AP2} = f(p^*) \geq (f(p_0) + f(p_1))/2 > \text{AP1}.$$

Hence, a two-period cycle is not optimal. Let us now consider a three-period cycle with prices p_0, p_1, and p_2 in the three periods. We arrive at 27 permutations using these prices. Proceeding as before, we find that when $\delta < \gamma$, charging p^* in all three periods results in an higher average profit per period during the cycle than any of the 27 permutations (the lengthy but straightforward proof is available from the authors).

Monopolist Retailer Managing Multiple Brands

LEMMA A1. *Let demand be specified by Equation (14) for brand i, $i = 1, \ldots, N$. Then, under an average profit per period criterion, if the brand prices follow a two period cycle, cycling in phase is a more profitable strategy than cycling out of phase.*

PROOF. Let N denote the number of brands, and suppose that the retailer maintains a two-period cyclical pricing policy for all of them. Let n_1 brands follow a hi-lo policy, and $n_2 = N - n_1$ brands follow a lo-hi policy; thus the prices of the n_2 brands are *out of phase* with those of n_1. Given this *out of phase* scenario, let p^*_{il} and p^*_{ih}, $i = 1, \ldots, N$, be the optimal lo and hi prices, respectively. The total profit during a two-period cycle is given by:

$$\begin{aligned}&\pi^*(n_1, n_2 \text{ out of phase})\\ &= \sum_{i=1}^{n_1} (p^*_{il} - c)\Big(a_i - b_i p^*_{il} + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p^*_{jl} + \sum_{k=1}^{n_2} \theta_{ik} p^*_{kh} + \delta_i(r_{il} - p^*_{il})\Big)\\ &\quad + \sum_{k=1}^{n_2} (p^*_{kh} - c)\Big(a_k - b_k p^*_{kh} + \sum_{i=1}^{n_1} \theta_{ki} p^*_{il} + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p^*_{mh} + \gamma_k(r_{kh} - p^*_{kh})\Big)\\ &\quad + \sum_{i=2}^{n_1} (p^*_{ih} - c)\Big(a_i - b_i p^*_{ih} + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p^*_{jh} + \sum_{k=1}^{n_2} \theta_{ik} p^*_{kl} + \gamma_i(r_{ih} - p^*_{ih})\Big)\\ &\quad + \sum_{k=1}^{n_2} (p^*_{kl} - c)\Big(a_k - b_k p^*_{kl} + \sum_{i=1}^{n_1} \theta_{ki} p^*_{ih} + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p^*_{ml} + \delta_k(r_{kl} - p^*_{kl})\Big).\end{aligned}$$

Employing the same prices, let the n_2 brands follow the two-period cyclical policy *in phase* with the n_1 brands. The total profit during a two-period cycle is given by:

$$\begin{aligned}&\pi(n_1, n_2 \text{ in phase})\\ &= \sum_{i=1}^{n_1} (p^*_{il} - c)\Big(a_i - b_i p^*_{il} + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p^*_{jl} + \sum_{k=1}^{n_2} \theta_{ik} p^*_{kl} + \delta_i(r_{il} - p^*_{il})\Big)\\ &\quad + \sum_{k=1}^{n_2} (p^*_{kl} - c)\Big(a_k - b_k p^*_{kl} + \sum_{i=1}^{n_1} \theta_{ki} p^*_{il} + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p^*_{ml} + \delta_k(r_{kl} - p^*_{kl})\Big)\\ &\quad + \sum_{i=2}^{n_1} (p^*_{ih} - c)\Big(a_i - b_i p^*_{ih} + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p^*_{jh} + \sum_{k=1}^{n_2} \theta_{ik} p^*_{kh} + \gamma_i(r_{ih} - p^*_{ih})\Big)\\ &\quad + \sum_{k=1}^{n_2} (p^*_{kh} - c)\Big(a_k - b_k p^*_{kh} + \sum_{i=1}^{n_1} \theta_{ki} p^*_{ih} + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p^*_{mh} + \delta_k(r_{kh} - p^*_{kh})\Big).\end{aligned}$$

Upon simplification,

$$\begin{aligned}&\pi(n_1, n_2 \text{ in phase}) - \pi(n_1, n_2 \text{ out of phase})\\ &= \sum_{i=1}^{n_1} \sum_{k=1}^{n_2} (p^*_{kh} - p^*_{kl})(p^*_{ih} - p^*_{il})(\theta_{il} + \theta_{ki}) > 0.\end{aligned}$$

Further, $\pi^*(n_1, n_2 \text{ in phase}) \geq \pi(n_1, n_2 \text{ out of phase})$; hence, the result.

PROPOSITION A1. i. *For $0 \leq \alpha < 1$, $0 < \beta < 1$, $\delta_i > \gamma_i$, $i = 1, \ldots, N$, if $f(p_{1t}, p_{2t}, \ldots, p_{Nt})$ is continuous and differentiable, and when the sum of discounted profits over an infinite horizon are maximized, a constant pricing policy for all the brands is not optimal. Therefore, at least one brand must have a cyclical pricing policy.*

ii. *For $0 \leq \alpha < 1$, $\delta_i > \gamma_i$, $i = 1, \ldots, N$, demand specified by Equation (14), and an average profit per period criterion, it is not optimal for any brand to maintain a constant price.*

iii. *Further, using an average profit per period criterion, if the N brands are symmetric (i.e., the respective parameter values for the N brands are equal) and if a hi-lo cycle is optimal, the hi prices of the N brands are equal, as are the lo prices*

PROOF OF PROPOSITION A1.i. The proof is by contradiction. Assume that each brand has a constant price p_k, $k = 1, \ldots, N$. Therefore $p_k(r_1, r_2, \ldots, r_N) = r_k$. Let $(r_{01}, r_{02}, \ldots, r_{0N}) = (r_1, r_2, \ldots, r_N)$ be the set of initial reference prices in an arbitrarily chosen period 0 and let $(p_{t1}, p_{t2}, \ldots, p_{tN})$ be the prices in period t. Thus, the reference price for brand k in period t is

$$r_{tk} = \alpha^t r_k + (1 - \alpha) \sum_{j=0}^{t-1} \alpha^{t-1-j} p_{jk}.$$

Consider a brand k. Since $p_k(r_1, r_2, \ldots, r_N) = r_k$, we have,

$$V(r_1, r_2, \ldots, r_k, \ldots, r_N) = \frac{f(r_1, r_2, \ldots, r_k, \ldots, r_N)}{1 - \beta}$$

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which must be larger than always charging $(r_1, r_2, \ldots, r_k + \Delta, \ldots, r_N)$ or $(r_1, r_2, \ldots, r_k - \Delta, \ldots, r_N)$, for $\Delta > 0$. Hence,

$$\frac{f(r_1, r_2, \ldots, r_N)}{1-\beta} \geq \frac{f(r_1, r_2, \ldots, r_k + \Delta, \ldots, r_N)}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \gamma_k (r_k + \Delta - c)(r_{tk} - (r_k + \Delta)),$$

$$\frac{f(r_1, r_2, \ldots, r_N)}{1-\beta} \geq \frac{f(r_1, r_2, \ldots, r_k + \Delta, \ldots, r_N)}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \delta_k (r_k - \Delta - c)(r_{tk} - (r_k - \Delta)).$$

Using the proof of Proposition 1, we arrive at a contradiction. Since brand k is arbitrarily chosen, the result holds for any brand. Hence, at least one brand should cycle.

PROOF OF PROPOSITION A1.ii. Let N denote the number of brands Let the retailer maintain constant prices, p_i^*, for $i = 1, \ldots, n_1$ brands and hi-lo prices p_{kh}^*, p_{kl}^* respectively for brands $k = 1, \ldots, n_2$; $n_2 = N - n_1$, where * indicates supremum prices for the corresponding class of pricing policy. We use this notation in the remainder of this appendix. We have,

$$r_{il} = \frac{p_{ih}^* + \alpha p_{il}^*}{1+\alpha}, \quad r_{ih} = \frac{p_{il}^* + \alpha p_{ih}^*}{1+\alpha}.$$

The total profit during a two-period cycle is given by:

$$\begin{aligned}
&\pi^*(n_1 \text{ constant}, n_2 \text{ cycle}) \\
&= \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kl}^*\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{kl}^* - c)\Big(a_k - b_k p_{kl}^* + \sum_{i=1}^{n_2} \theta_{ki} p_i^* \\
&\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{ml}^* + \delta_k (r_{kl} - p_{kl}^*)\Big) \\
&\quad + \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kh}^*\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{kh}^* - c)\Big(a_k - b_k p_{kh}^* + \sum_{i=1}^{n_1} \theta_{ik} p_i^* \\
&\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{mh}^* + \gamma_k (r_{kh} - p_{kh}^*)\Big).
\end{aligned}$$

Now, let the n_1 brands also follow a cyclical policy, i.e., $p_{il} = (p_i^* + e)$ and $p_{ih} = (p_i^* - e)$. The total profit during a two-period cycle is given by:

$$\begin{aligned}
&\pi(n_1 \text{ cycle}, n_2 \text{ cycle}) \\
&= \sum_{i=1}^{n_1} (p_i^* - e - c)\Big(a_i - b_i (p_i^* - e) + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} (p_j^* - e) \\
&\qquad + \sum_{k=1}^{n_2} \theta_{ik} p_{kl}^* + \delta_i (r_{il} - p_{il})\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{kl}^* - c)\Big(a_k - b_k p_{kl}^* + \sum_{i=1}^{n_1} \theta_{ki} (p_i^* - e) \\
&\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{ml}^* + \delta_k (r_{kl} - p_{kl}^*)\Big) \\
&\quad + \sum_{i=1}^{n_1} (p_i^* + e - c)\Big(a_i - b_i (p_i^* + e) + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} (p_j^* + e) \\
&\qquad + \sum_{k=1}^{n_2} \theta_{ik} p_{kh}^* + \gamma_i (r_{ih} - p_{ih})\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{kh}^* - c)\Big(a_k - b_k p_{kh}^* + \sum_{i=1}^{n_1} \theta_{ki} (p_i^* + e) \\
&\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{mh}^* + \gamma_k (r_{kh} - p_{kh}^*)\Big).
\end{aligned}$$

Upon simplification, we find that $\exists e < e^*$: $\pi(n_1 \text{ cycle}, n_2 \text{ cycle}) - \pi^*(n_1 \text{ constant}, n_2 \text{ cycle}) > 0$, and

$$e^* = \frac{\Sigma_{i=1}^{n_1} \left\{ \dfrac{2(p_i^* - c)(\delta_i - \gamma_i)}{1+\alpha} + \Sigma_{k=1}^{n_2} (p_{kh}^* - p_{kl}^*)(\theta_{ik} + \theta_{ki}) \right\}}{2\, \Sigma_{i=1}^{n_1} \left\{ \dfrac{\delta_i + \gamma_i}{1+\alpha} + b_i - \Sigma_{j=1, j\neq i}^{n_1} \theta_{ij} \right\}} > 0.$$

The above result holds for $n_1, n_2 > 0$, $\delta_i > \gamma_i$; therefore, when $\delta_i > \gamma_i$, given the alternatives of hi-lo and constant pricing, it is more profitable to maintain hi-lo prices for all brands.

PROOF OF PROPOSITION A1.iii. This result is established by setting the corresponding parameter values across all brands to be equal and considering the profit during a hi-lo pricing cycle. Solving the simultaneous system of equations obtained from the first order conditions provides the result.

PROPOSITION A2. i. *For $0 \leq \alpha < 1$, $\delta_i \leq \gamma_i$, $i = 1, \ldots, N$, demand specified by Equation (14), and an average profit per period criterion, charging a constant price for all the brands is more profitable than charging hi-lo prices for any brand*

ii. *Further, using an average profit per period criterion, if the N brands are symmetric and if a constant policy is optimal, the prices of the N brands are equal.*

PROOF OF PROPOSITION A2.i. Let N denote the number of brands. Let the retailer maintain constant prices, p_i^*, for brands $i = 1, \ldots, n_1$ and hi-lo prices p_{kh}^*, p_{kl}^*, respectively, for brands $k = 1, \ldots, n_2$, $n_2 = N - n_1$. The total profit during a 2 period cycle is given by:

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$$\begin{aligned}
&\pi^*(n_1 \text{ constant}, n_2 \text{ cycle}) \\
&= \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kl}^*\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{kl}^* - c)\Big(a_k - b_k p_{kl}^* + \sum_{i=1}^{n_1} \theta_{ki} p_i^* \\
&\qquad\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{ml}^* + \delta_k (r_{kl} - p_{kl}^*)\Big) \\
&\quad + \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kh}^*\Big) \\
&\quad + \sum_{k=1}^{n_2} (p_{k=1}^* - c)\Big(a_k - b_k p_{kh}^* + \sum_{i=1}^{n_1} \theta_{ki} p_i^* \\
&\qquad\qquad + \sum_{m=1, m\neq k}^{n_2} \theta_{km} p_{mh}^* + \gamma_k (r_{kh} - p_{kh}^*)\Big).
\end{aligned}$$

Now let us consider a constant pricing policy, p_n, for brand n, $1 \leq n \leq n_2$. The total profit in two periods is given by:

$$\begin{aligned}
&\pi(n_1 \text{ constant}, n_2 - 1 \text{ cycle, brand } n \text{ constant}) \\
&= \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2 - 1} \theta_{ik} p_{kl}^* + \theta_{in} p_n\Big) \\
&\quad + \sum_{k=1}^{n_2 - 1} (p_{kl}^* - c)\Big(a_k - b_k p_{kl}^* + \sum_{i=1}^{n_1} \theta_{ki} p_i^* \\
&\qquad\qquad + \sum_{m=1, m\neq k}^{n_2 - 1} \theta_{km} p_{ml}^* + \theta_{kn} p_n + \delta_k (r_{kl} - p_{kl}^*)\Big) \\
&\quad + (p_n - c)\Big(a_n - b_n p_n + \sum_{i=1}^{n_1} \theta_{in} p_i^* + \sum_{k=1}^{n_2 - 1} \theta_{nk} p_{kl}^*\Big) \\
&\quad + \sum_{i=1}^{n_1} (p_i^* - c)\Big(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2 - 1} \theta_{ik} p_{kh}^* + \theta_{in} p_n\Big) \\
&\quad + \sum_{k=1}^{n_2 - 1} (p_{kh}^* - c)\Big(a_k - b_k p_{kh}^* + \sum_{i=1}^{n_1} \theta_{ki} p_i^* \\
&\qquad\qquad + \sum_{m=1, m\neq k}^{n_2 - 1} \theta_{km} p_{mh}^* + \theta_{kn} p_n + \gamma_k (r_{kh} - p_{kh}^*)\Big) \\
&\quad + (p_n - c)\Big(a_n - b_n p_n + \sum_{i=1}^{n_1} \theta_{in} p_i^* + \sum_{k=1}^{n_2 - 1} \theta_{nk} p_{kh}^*\Big).
\end{aligned}$$

Upon simplification, we find that a sufficient condition for $\pi(n_1$ constant, $(n_2 - 1)$ cycle, brand n constant$) > \pi^*(n_1$ constant, n_2 cycle$)$ is that

$$\exists p_n > p_n^* = \frac{p_{nl} \sum_{k=1}^{n_2 - 1} \theta_{nk} p_{kl}^* + p_{nh}^* \sum_{k=1}^{n_2 - 1} \theta_{nk} p_{kh}^*}{\sum_{k=1}^{n_2 - 1} \theta_{nk} (p_{kl}^* + p_{kh}^*)}.$$

Note that p_n^* is a weighted average of p_{nl}^* and p_{nh}^*, thus showing that the price range is feasible. The above holds for $n_1, n_2 > 0$, $\delta_i < \gamma_i$; therefore, when $\delta_i < \gamma_i$, given the alternatives of hi-lo and constant pricing, it is more profitable to maintain constant prices for all the brands.

PROOF OF PROPOSITION A2.ii. The approach parallels that for Proposition A1.iii.

PROPOSITION A3. *For $0 \leq \alpha < 1$, $\delta_i > \gamma_i$, $i = 1, \ldots, n_1$; $\delta_j \leq \gamma_j$, $j = 1, \ldots, N - n_1$, demand specified by Equation (14), and an average profit per period criterion, it is not optimal to charge a constant price for any of the n_1 brands, and it is more profitable to maintain constant prices than hi-lo prices for the $N - n_1$ brands.*

PROOF. Consider N brands with $\delta_i > \gamma_i$, $i = 1, \ldots, n_1$, and $\delta_j < \gamma_j$, $j = 1, \ldots, n_2$, where $n_2 = N - n_1$. Let n_{11} brands follow a hi-lo pricing policy and $n_{12} = n_1 - n_{11}$ brands follow a constant pricing policy. Similarly, let n_{21} brands follow a hi-lo pricing policy and $n_{22} = n_2 - n_{21}$ brands follow a constant pricing policy. Using the procedures outlined for Propositions A1.ii and A2.i, we show that the result holds.

Duopoly

PROOF OF LEMMA 1. Let the subscripts $i = 1, 2$ index the firms and l and h denote lo and hi prices, respectively. Let the two brands maintain hi-lo prices *out of phase*. Given this *out of phase* scenario, let p_{il}^* and p_{ih}^* be the optimal lo and hi prices. The total profit for firm i during a two-period cycle when prices in the are out of phase is given by:

$$\begin{aligned}
&\pi_i^*(\text{out of phase}) \\
&= (p_{il}^* - c)(a_i - b_i p_{il}^* + \theta_{ij} p_{jh}^* + \delta_i (r_{il} - p_{il}^*)) \\
&\quad + (p_{il}^* - c)(a_i - b_i p_{ih}^* + \theta_{ij} p_{jl}^* + \gamma_i (r_{ih} - p_{il}^*)) \\
&= (p_{il}^* - c)(a_i - b_i p_{il}^*) + \theta_{ij} p_{jh} (p_{il}^* - c) + \delta_i (p_{il}^* - c) \frac{p_{ih}^* - p_{il}^*}{1 + \alpha} \\
&\quad + (p_{ih}^* - c)(a_i - b_i p_{ih}^*) + \theta_{ij} p_{jl}^* (p_{ih}^* - c) + \gamma_i (p_{ih}^* - c) \frac{p_{il}^* - p_{ih}^*}{1 + \alpha}.
\end{aligned}$$

Now, utilizing the same prices, let brand i follow the 2 period cyclical policy *in phase* with brand j. The total profit for firm i during a 2 period cycle is given by:

$$\begin{aligned}
&\pi_i(\text{in phase}) \\
&= (p_{il}^* - c)(a_i - b_i p_{il}^* + \theta_{ij} p_{jl}^* + \delta_i (r_{il} - p_{il}^*)) \\
&\quad + (p_{ih}^* - c)(a_i - b_i p_{ih}^* + \theta_{ij} p_{jl}^* + \gamma_i (r_{ih} - p_{il}^*)) \\
&= (p_{il}^* - c)(a_i - b_i p_{il}^*) + \theta_{ij} p_{jh} (p_{il}^* - c) + \delta_i (p_{il}^* - c) \frac{p_{ih}^* - p_{il}^*}{1 + \alpha} \\
&\quad + (p_{ih}^* - c)(a_i - b_i p_{ih}^*) + \theta_{ij} p_{jh}^* (p_{ih}^* - c) + \gamma_i (p_{ih}^* - c) \frac{p_{il}^* - p_{ih}^*}{1 + \alpha},
\end{aligned}$$

$$\begin{aligned}
&\pi_i(\text{in phase}) - \pi_i^*(\text{out of phase}) \\
&= (p_{il}^* - c)\theta_{ij}(p_{jl}^* - p_{jh}^*) + (p_{ih}^* - c)\theta_{ij}(p_{jh}^* - p_{jl}^*) \\
&= \theta_{ij}(p_{jh}^* - p_{jl}^*)(p_{ih}^* - c - p_{il}^* + c) > 0.
\end{aligned}$$

Further, $\pi_i^*(\text{in phase}) \geq \pi_i(\text{in phase})$; hence, the result.

PROOF OF PROPOSITION 3.i. Among the class of constant prices, assume firms i and j choose prices, p_i^* and p_j^*, respectively. We will show that $\exists e_i$: charging $p_{il} = p_i^* - e_i$ and $p_{ih} = p_i^* + e_i$ produces a higher profit per period for firm i. The total profit when firm i has cyclical prices and firm j has a constant price is given by:

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$$\begin{aligned}
&\pi_i(i \text{ cycles}, j \text{ constant at } p_j^*)\\
&= (p_{il} - c)(a_i - b_i p_{il} + \theta_{ij} p_j^* + \delta_i(r_{il} - p_{il}))\\
&\quad + (p_{ih} - c)(a_i - b_i p_{ih} + \theta_{ij} p_j^* + \gamma_i(r_{ih} - p_{ih}))\\
&= (p_i^* - c - e_i)(a_i - b_i p_i^* + \theta_{ij} p_j^* + b_i e_i + \delta_i(r_{il} - p_{il}))\\
&\quad + (p_i^* - c + e_i)(a_i - b_i p_i^* + \theta_{ij} p_j^* - b_i e_i + \gamma_i(r_{ih} - p_{ih}))\\
&= 2(p_i^* - c)(a_i - b_i p_i^* + \theta_{ij} p_j^*) + b_i e_i (p_i^* - c)\\
&\quad + (p_i^* - c)\delta_i(r_{il} - p_{il}) - e_i(a_i - b_i p_i^* + \theta_{ij} p_j^*)\\
&\quad - b_i e_i^2 - e_i \delta(r_{il} - p_{il}) - b_i e_i (p_i^* - c) + (p_i^* - c)\gamma_i(r_{ih} - p_{ih})\\
&\quad + e_i(a_i - b_i p_i^* + \theta_{ij} p_j^*)\\
&\quad - b_i e_i^2 + e_i \gamma_i (r_{ih} - p_{ih}),\\
&= \pi_i^*(i \text{ constant at } p_i^*, j \text{ constant at } p_j^*)\\
&\quad + (p_i^* - c)(\delta_i(r_{il} - p_{il}) + \gamma_i(r_{ih} - p_{ih}))\\
&\quad - 2b_i e_i^2 - e_i(\delta_i(r_{il} - p_{il}) - \gamma_i(r_{ih} - p_{ih}))\\
&= \pi_i^*(i \text{ constant at } p_i^*, j \text{ constant at } p_j^*)\\
&\quad + \frac{2e_i}{1+\alpha}((p_i^* - c)(\delta_i - \gamma_i) - e_i(\delta_i + \gamma_i + b_i(\alpha + 1))).
\end{aligned}$$

Thus,

$$\pi_i(i \text{ cycles}, j \text{ constant at } p_j^*) > \pi_i^*(i \text{ constant at } p_i^*, j \text{ constant at } p_j^*),$$

$$\text{for all } e_i \cdot e_i < e_i^*, \quad \text{where } e_i^* = \frac{(p_i^* - c)(\delta_i - \gamma_i)}{\delta_i + \gamma_i + b_i(1+\alpha)}.$$

Hence, when firm j charges a constant price, charging a constant price is not a best response for firm i.

Among the class of two-period cyclical prices, let firm j choose prices p_{jl}^* and p_{jh}^*. Among the class of constant prices, let p_i^* be the supremum for firm i. We will show that $\exists k_i$: charging $p_{il} = p_i^* - k_i$, and $p_{ih} = p_i^* + k_i$, produces a higher per period profit for firm i. The total profit for firm i when i has a constant price and j has cyclical prices is given by:

$$\begin{aligned}
&\pi_i^*(i \text{ constant at } p_i^*, j \text{ cycles})\\
&= (p_i^* - c)(a_i - b_i p_i^* + \theta_{ij} p_{jl}^*) + (p_i^* - c)(a_i - b_i p_i^* + \theta_{ij} p_{jh}^*)\\
&= (p_i^* - c)(2a_i - 2b_i p_i^* + \theta_{ij}(p_{jl}^* + p_{jh}^*)).
\end{aligned}$$

The total profit for firm i during a two-period cycle when both firms have cyclical prices is given by:

$$\begin{aligned}
&\pi_i(i \text{ cycles}, j \text{ cycles})\\
&= (p_i^* - c - k_i)(a_i - b_i p_i^* + \theta_{ij} p_{jl}^* + b_i k_i + \delta_i(r_{il} - p_{il}))\\
&\quad + (p_i^* - c + k_i)(a_i - b_i p_i^* + \theta_{ij} p_{jh}^* - b_i k_i + \gamma_i(r_{ih} - p_{ih})).
\end{aligned}$$

Upon simplification, we find that $\exists k_i : k_i < k_i^*$, where

$$\pi_i(i \text{ cycles}, j \text{ cycles}) > \pi_i^*(i \text{ constant at } p_i^*, j \text{ cycles})$$

and

$$k^* = (p_i^* - c)\frac{\delta_i - \gamma_i}{\delta_i + \gamma_i + b_i(1+\alpha)}.$$

Hence, when firm j has cyclical prices, charging a constant price is not a best response for firm i.

We have shown that when $\delta_i > \gamma_i$, $\pi_i(i \text{ cycles}, j \text{ constant}) > \pi_i(i \text{ constant at } p_i^*, j \text{ constant at } p_j^*)$ and $\pi_i(i \text{ cycles}, j \text{ cycles}) > \pi_i(i \text{ constant at } p_i^*, j \text{ cycles})$. Thus, the MPE Nash solution is not to have constant prices. Therefore in a duopoly, prices cycle in phase.

PROOF OF PROPOSITION 3.ii. The approach parallels that for Proposition A1.iii.

PROOF OF PROPOSITION 4.i. Here we show that when $\delta_i < \gamma_i$, between the alternatives of charging a constant price or cyclical prices with period 2, the equilibrium solution is charging a constant price. Let firm i choose a hi-lo pricing policy such that p_{il}^* and p_{ih}^* are the respective lo and hi prices. Now,

$$\begin{aligned}
&\pi_i^*(i \text{ cycles}, j \text{ constant at } p_j^*)\\
&= (p_{il}^* - c)(a_i - b_i p_{il}^* + \theta_{ij} p_j^* + \delta_i(r_{il} - p_{il}^*))\\
&\quad + (p_{ih}^* - c)(a_i - b_i p_{ih}^* + \theta_{ij} p_j^* + \gamma_i(r_{ih} - p_{ih}^*)).
\end{aligned}$$

Similarly,

$$\begin{aligned}
&\pi_i(i \text{ constant at } p_i, j \text{ constant at } p_j^*)\\
&= (p_i - c)(a_i - b_i p_i + \theta_{ij} p_j^*) + (p_i - c)(a_i - b_i p_i + \theta_{ij} p_j^*).
\end{aligned}$$

Upon simplification, we find that a sufficient condition for $\pi_i(i \text{ constant at } p_i, j \text{ constant at } p_j^*) > \pi_i^*(i \text{ cycles}, j \text{ constant at } p_j^*)$ is that

$$\exists p_i : \frac{p_{il}^* + p_{ih}^*}{2} < p_i < \left(\frac{(p_{il}^*)^2 + (p_{ih}^*)^2}{2}\right)^{1/2}.$$

Note that the lower bound for p_i is the average of the lo and hi prices. Thus, if firm j has a constant price, maintaining a constant price will result in a higher profit per period for firm i than having a two-period pricing cycle.

Now let us consider the case where firm j maintains cyclical prices. Of the class of two-period pricing cycles let p_{jl}^* and p_{jh}^* be the supremum. Similarly, let firm i choose two prices p_{il}^* and p_{ih}^*. The total profit for firm i in this scenario is given by:

$$\begin{aligned}
\pi_i^*(i \text{ cycles}, j \text{ cycles}) &= (p_{il}^* - c)(a_i - b_i p_{il}^* + \theta_{ij} p_{jl}^* + \delta_i(r_{il} - p_{il}^*))\\
&\quad + (p_{ih}^* - c)(a_i - b_i p_{ih}^* + \theta_{ij} p_{jh}^* + \gamma_i(r_{ih} - p_{ih}^*)).
\end{aligned}$$

Similarly,

$$\begin{aligned}
&\pi_i(i \text{ constant at } p_i, j \text{ cycles})\\
&= (p_i - c)(a_i - b_i p_i + \theta_{ij} p_{jl}^*) + (p_i - c)(a_i - b_i p_{ih} + \theta_{ij} p_{jh}^*).
\end{aligned}$$

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Upon simplification, we find that a sufficient condition for $\pi_i(i$ constant at p_i, j cycles$) > \pi_i^*(i$ cycles, j cycles$)$ is that

$$\exists p_i.\ \min\left\{\frac{p_{il}^* + p_{ih}^*}{2}, \frac{p_{il}^* p_{jl}^* + p_{ih}^* p_{jh}^*}{p_{jl}^* + p_{jh}^*}\right\} \le p_i \le \left(\frac{(p_{il}^*)^2 + (p_{ih}^*)^2}{2}\right)^{1/2}.$$

Note that the lower bound for p_i is a weighted average of the lo and hi prices. Hence, when firm j has cyclical prices, charging a constant price yields higher profits for firm i than maintaining hi-lo prices. Thus, we have shown that when $\delta_i < \gamma_i$, $\pi_i(i$ constant at p_i^*, j constant at $p_j^*) > \pi_i(i$ cycles, j constant at $p_j^*)$ and $\pi_i(i$ constant at p_i^*, j cycles$) > \pi_i(i$ cycles, j cycles$)$. Therefore, the MPE Nash solution is to choose a constant price over hi-lo prices.

PROOF OF PROPOSITION 4.ii. The approach parallels that for Proposition A1.iii.

PROOF OF PROPOSITION 5. This result is obtained using the approach outlined for Proposition A3.

Oligopoly

LEMMA A2. *Let demand be specified by Equation (14) for brand i, $i = 1, \ldots, N$. Then, under an average profit per period criterion, if the brand prices follow a two period cycle in phase, there is no incentive for brand i, $i = 1, \ldots, N$, to cycle out of phase; hence a MPE Nash solution is to maintain the cyclical prices in phase.*

PROOF. Let the N brands in an oligopoly maintain a two-period cyclical pricing policy. Let $(N-1)$ brands follow a hi-lo policy and let the Nth brand follow a hi-lo policy; thus the prices of the Nth brand are *out of phase* with the other $(N-1)$ brands Given this out of phase scenario, let p_{il}^* and p_{ih}^*, $i = 1, \ldots, N$, be the optimal lo and hi prices. The total profit for brand N during a two-period cycle is given by:

$$\pi_N^*(i \text{ out of phase}) = (p_{ih}^* - c)\left(a_i - b_i p_{ih}^* + \sum_{j=1, j\neq i}^{N} \theta_{ij} p_{jl}^* + \gamma_i(r_{ih} - p_{ih}^*)\right) + (p_{il}^* - c)\left(a_i - b_i p_{il}^* + \sum_{j=1, j\neq i}^{N} \theta_{ij} p_{jh}^* + \delta_i(r_{ih} - p_{ih}^*)\right).$$

Now, using the same prices, let brand N follow the two-period cyclical policy *in phase* with the other $(N-1)$ brands. The total profit for brand N during a two-period cycle is given by:

$$\pi_N(\text{in phase}) = (p_{il}^* - c)\left(a_i - b_i p_{ie}^* + \sum_{j=1, j\neq i}^{N} \theta_{ij} p_{jl}^* + \delta_i(r_{il} - p_{il}^*)\right) + (p_{ih}^* - c)\left(a_i - b_i p_{ih}^* + \sum_{j=1, j\neq i}^{N} \theta_{ij} p_{jh}^* + \gamma_i(r_{ih} - p_{ih}^*)\right).$$

Upon simplification,

$$\pi_i(\text{in phase}) - \pi_i^*(i \text{ out of phase}) = \sum_{j=1, j\neq i}^{N} (p_{jh}^* - p_{jl}^*)(p_{ih}^* - p_{il}^*)\theta_{ij} > 0.$$

Further, $\pi_i^*(\text{in phase}) \ge \pi_i(\text{in phase})$; hence, the result.

PROPOSITION A4. i. *For $0 \le \alpha < 1$, $\delta_i > \gamma_i$, $i = 1, \ldots, N$, demand specified by Equation (14), and an average profit per period criterion, the MPE Nash solution is not to have constant prices for any one or more of the N brands. Hence, the equilibrium policy cycles.*

ii. *Further, using an average profit per period criterion, if a hi-lo cycle is optimal and if the N brands are symmetric, the hi prices of all the N brands are equal, as are the lo prices.*

PROOF OF PROPOSITION A4.i. Let N denote the number of brands in the oligopoly. Let n_1 brands maintain constant prices p_i^*, $i = 1, \ldots, N$ and let $n_2 = N - n_1$ brands maintain hi-lo prices p_{kl}^* and p_{kh}^*, $k = 1, \ldots, n_2$. Consider a brand i, $1 \le i \le n_1$; we show that this firm has an incentive to move to a cyclical policy. Hence, in equilibrium no brand in the oligopoly will have a constant price when $\delta_i > \gamma_i$. The total profit for i in two periods is given by:

$$\pi_i^*(n_1 \text{ constant}, n_2 \text{ cycle}) = (p_i^* - c)\left(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kl}^*\right) + (p_i^* - c)\left(a_i - b_i p_i^* + \sum_{j=1, j\neq i}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kh}^*\right).$$

Now, let the brand i consider hi-lo prices, i.e., $p_{il} = p_i^* + e_i$ and $p_{ih} = p_i^* - e_i$, respectively. The total profit for i in two periods is given by,

$$\pi_i(i \text{ cycles}, n_1 - 1 \text{ constant}, n_2 \text{ cycle}) = (p_i^* - e_i - c)\left(a_i - b_i(p_i^* - e_i) + \sum_{j=1}^{n_1-1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kl}^* + \delta_i(r_{il} - p_{il})\right) + (p_i^* + e_i - c)\left(a_i - b_i(p_i^* + e_i) + \sum_{j=1}^{n_1-1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2} \theta_{ik} p_{kh}^* + \gamma_i(r_{ih} - p_{ih})\right).$$

Upon simplification, we find that a sufficient condition for $\pi_i(i$ cycles, $n_1 - 1$ constant, n_2 constant$) > \pi_i(n_1$ constant, n_2 cycle$)$ is that

$$\exists e_i : e_i < e_i^* = \frac{1}{2(b_i(1+\alpha) + \delta_i + \gamma_i)} \times \left\{2(p_i^* - c)(\delta_i - \gamma_i) + (1+\alpha)\sum_{k=1}^{n_2} \theta_{ik}(p_{kh}^* - p_{kl}^*)\right\}.$$

Since firm i, $1 \le i \le n_1$ is arbitrary, the MPE Nash strategy in an oligopoly is not to maintain constant prices when $\delta_i > \gamma_i$.

PROOF OF PROPOSITION A4.ii. The approach parallels that for Proposition A1.iii.

PROPOSITION A5. i. *For $0 \le \alpha < 1$, $\delta_i \le \gamma_i$, $i = 1, N$, demand specified by Equation (14), and an average profit per period profit criterion, between the option of either charging a constant price or hi-lo prices, the MPE Nash solution is to maintain a constant price.*

ii *Further, using an average profit per period criterion, if a constant policy is optimal, and if the N brands are symmetric, the prices of the N brands are equal.*

PROOF OF PROPOSITION A5.i. Let N denote the number of brands in the oligopoly. Let n_1 brands maintain constant prices p_i^*, $i = 1, \ldots, N$ and let $n_2 = N - n_1$ brands maintain hi-lo prices p_{kl}^* and p_{kh}^*, $k = 1,$

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$\ldots, n_2$. Consider a brand i, $1 \le i \le n_2$; we show that this firm has an incentive to move to a constant policy. Hence, in equilibrium no brand in the oligopoly will have hi-lo prices when $\delta_i < \gamma_i$. The total profit for i in two periods is given by

$$\pi_i^*(n_1 \text{ constant}, n_2 \text{ cycle})$$
$$= (p_{il}^* - c)\left(a_i - b_i p_{il}^* + \sum_{j=1}^{n_1} \theta_{ij} p_j^* + \sum_{k=1, k\neq i}^{n_2} \theta_{ik} p_{kl}^* + \delta_i(r_{il} - p_{il}^*)\right)$$
$$+ (p_{ih}^* - c)\left(a_i - b_i p_{ih}^* + \sum_{j=1}^{n_1} \theta_{ij} p_j^* + \sum_{k=1, k\neq i}^{n_2} \theta_{ik} p_{kh}^* + \gamma_i(r_{ik} - p_{ih}^*)\right).$$

Now, let firm i consider a constant price, p_i. The total profit for i in two periods is given by

$$\pi_i(i \text{ constant}, n_1 \text{ constant}, n_2 - 1 \text{ cycle})$$
$$= (p_i - c)\left(a_i - b_i p_i + \sum_{j=1}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2-1} \theta_{ik} p_{kl}^*\right)$$
$$+ (p_i - c)\left(a_i - b_i p_i + \sum_{j=1}^{n_1} \theta_{ij} p_j^* + \sum_{k=1}^{n_2-1} \theta_{ik} p_{kh}^*\right).$$

Upon simplification, we find that a sufficient condition for $\pi_i(i$ constant at p_i, n_1 constant, $n_2 - 1$ cycle$) > \pi_i^*(n_1$ constant, $n_2 - 1$ cycle$)$ is that

$$\exists p_i\colon \min\left\{\frac{p_{il}^* + p_{ih}^*}{2}, \frac{p_{il}^* \sum_{k=1}^{n_2-1} \theta_{ik} p_{kl}^* + p_{ih}^* \sum_{k=1}^{n_2-1} \theta_{ik} p_{kh}^*}{\sum_{k=1}^{n_2-1} \theta_{ik}(p_{kl}^* + p_{kh}^*)}\right.$$
$$\le p_i \le \left(\frac{(p_{il}^*)^2 + (p_{ih}^*)^2}{2}\right)^{1/2}.$$

Note that the lower bound for p_i is a weighted average of the lo and hi prices. Thus, when $\delta_i < \gamma_i$, there is an incentive for brand i to charge a constant instead of hi-lo prices. Since firm i, $1 \le i \le n_2$ is arbitrarily chosen, given the alternatives of offering a constant price or hi-lo prices, the MPE Nash strategy in an oligopoly is to maintain constant prices when $\delta_i < \gamma_i$.

PROOF OF PROPOSITION A5.ii. The approach parallels that for Proposition A1.iii.

PROPOSITION A6. *For $0 \le \alpha < 1$, $\delta_i > \gamma_i$, $i = 1, \ldots, n_1$, $\delta_j \le \gamma_j$, $j = 1, \ldots, N - n_1$, demand specified by Equation (14), and an average profit per period criterion, the equilibrium solution is not to charge a constant price for brand i, $1 \le i \le n_1$, and between the option of charging a constant price or maintaining hi-lo prices, the equilibrium solution is to maintain a constant price for brand j, $1 \le j \le N - n_1$.*

PROOF. This result is obtained using the approach outlined for Proposition A3. □

Impact of Heterogeneity

PROOF OF PROPOSITION 6. Retailer Managing N brands. For exposition, we only provide the proof for 2 brands which easily extends to the case of N brands. Let subscripts 1 and 2 denote consumer segments, A and B denote brands. Let q_1 denote share of segment 1. Therefore, share of segment 2 is $1 - q_1$. We have $\delta_1 > \gamma_1$ and $\delta_2 < \gamma_2$.

Of the class of constant policies, let the retailer charge constant prices p_A^* and p_B^*. The total profit in 2 periods is given by:

$$\pi(\text{constant})$$
$$= 2[(p_A^* - c)(a - bp_A^* + \theta p_B^*) + (p_B^* - c)(a - bp_B^* + \theta p_A^*)].$$

Consider hi-lo prices such that for brand A, $p_{hA} = p_A^* + k_A$ and $p_{lA} = p_A^* - k_A$. Similarly, consider hi-lo prices for brand B. The total profit during a 2 period hi-lo cycle in phase is given by:

$$\pi(\text{cycle})$$
$$= (p_A^* + k_A - c)[q_1(a - bp_A^* - bk_A + \theta(p_B^* + k_B) + \gamma_1(r_{lA1} - p_{hA}))$$
$$+ (1 - q_1)(a - bp_A^* - bk_A + \theta(p_B^* + k_B) + \gamma_2(r_{lA2} - p_{hA}))]$$
$$+ (p_B^* + k_B - c)[q_1(a - bp_B^* - bk_B + \theta(p_A^* + k_A) + \gamma_1(r_{lB1} - p_{hB}))$$
$$+ (1 - q_1)(a - bp_B^* - bk_B + \theta(p_A^* + k_A) + \gamma_2(r_{lB2} - p_{hB}))]$$
$$+ (p_A^* + k_A - c)[q_1(a - bp_A^* + bk_A + \theta(p_B^* + k_B) + \delta_1(r_{hA1} - p_{lA}))$$
$$+ (1 - q_1)(a - bp_A^* - bk_A + \theta(p_B^* - k_B) + \delta_2(r_{hA2} - p_{lA}))]$$
$$+ (p_B^* + k_B - c)[q_1(a - bp_B^* + bk_B + \theta(p_A^* + k_A) + \delta_1(r_{hB1} - p_{lB}))$$
$$+ (1 - q_1)(a - bp_B^* - bk_B + \theta(p_A^* - k_A) - \delta_2(r_{hB2} - p_{lB}))],$$

where

$$r_{hi\mathrm{A(B)}} = \frac{p_{h\mathrm{A(B)}} + \alpha_i p_{la}}{1 + \alpha_j}, \quad r_{li\mathrm{A(B)}} = \frac{p_{l\mathrm{A(B)}} + \alpha_i p_{ha}}{1 + \alpha_i}, \qquad i = 1, 2.$$

Upon simplification, we find that $k_A = k_b = k > 0$ will exist such that

$$\pi(\text{cycle}) > \pi(\text{constant}) \quad \text{if} \quad \left(\frac{q_1}{1 - q_1}\right)\left(\frac{\delta_1 - \gamma_1}{\gamma_2 - \delta_2}\right)$$
$$> \left(\frac{1 + \alpha_1}{1 + \alpha_2}\right).$$

Duopoly/Oligopoly. Let firm B follow a constant price Of the class of constant prices, let p_B^* be the optimal price. If firm A also follows a constant price p_A^*, the total profit for firm A during 2 periods is given by:

$$\pi_A(\text{A constant, B constant}) = 2(p_A^* - c)(a - bp_A^* + \theta p_B^*).$$

Let firm A consider hi-lo prices, p_{hA} and p_{lA} such that $p_{hA} = p_A^* + k_A$ and $p_{lA} = p_A^* - k_A$. The total profit during a two-period cycle is given by:

$$\pi_A(\text{A cycle, B constant})$$
$$= (p_A^* + k_A - c)[q_1(a - bp_A^* - bk_A + \theta p_B^* + \gamma_1(r_{lA1} - p_{hA}))$$
$$+ (1 - q_1)(a - bp_A^* - bk_A + \theta p_B^* + \gamma_2(r_{lA2} - p_{hA}))]$$
$$+ (p_A^* - k_A - c)[q_1(a - bp_A^* + bk_A + \theta p_B^* + \delta_1(r_{hA1} - p_{hA}))$$
$$+ (1 - q_1)(a - bp_A^* - bk_A + \theta p_B^* + \delta_2(r_{hA2} - p_{lA}))].$$

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Upon simplification, we find that $k_A > 0$ will exist such that:

π_A(A cycle, B constant) > π_A(A constant, B constant)

$$\text{if } \left(\frac{q_1}{1-q_1}\right)\left(\frac{\delta_1-\gamma_1}{\gamma_2-\delta_2}\right) > \left(\frac{1+\alpha_1}{1+\alpha_2}\right).$$

Let firm B follow hi-lo prices Of the class of two-period cyclical prices, let p^*_{hB} and p^*_{lB} be the optimal prices. If firm A follows a constant price p^*_A, the total profit for firm A during 2 periods is given by:

$$\pi_A(\text{A constant, B cycle}) = (p^*_A - c)(2a - 2bp^*_A + \theta(p^*_{hB} + p^*_{lB})).$$

Let firm A consider hi-lo prices, p_{hA} and p_{lA} such that $p_{hA} = p^*_A + k_A$ and $p_{lA} = p^*_A - k_A$. The total profit during a two-period cycle is given by:

$$\begin{aligned}
&\pi_A(\text{A cycle, B cycle})\\
&\quad = (p^*_A + k_A - c)[q_1(a - bp^*_A - bk_A + \theta p^*_{hB} + \gamma_1(r_{lA1} - p_{hA}))\\
&\quad + (1-q_1)(a - bp^*_A - bk_A + \theta p^*_{hB} + \gamma_2(r_{lA2} - p_{hA}))]\\
&\quad + (p^*_A - k_A - c)[q_1(a - bp^*_A + bk_A + \theta p^*_{lB} + \delta_1(r_{hA1} - p_{hA}))\\
&\quad + (1-q_1)(a - bp^*_A - bk_A + \theta p^*_{lB} + \delta_2(r_{hA2} - p_{lA}))].
\end{aligned}$$

Upon simplification, we find that $k_A > 0$ will exist such that:

π_A(A cycle, B cycle) > π_A(A constant, B cycle)

$$\text{if } \left(\frac{q_1}{1-q_1}\right)\left(\frac{\delta_1-\gamma_1}{\gamma_2-\delta_2}\right) > \left(\frac{1+\alpha_1}{1+\alpha_2}\right).$$

Thus, in a duopoly, maintaining hi-lo prices will be a MPE Nash equilibrium solution if

$$\frac{q_1}{1-q_1}\frac{\delta_1-\gamma_1}{\gamma_2-\delta_2} > \frac{1+\alpha_1}{1+\alpha_2}.$$

It is straightforward to extend this proof to an oligopoly with N symmetric brands. □

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